

Yellow River Marsh Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 09 April, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

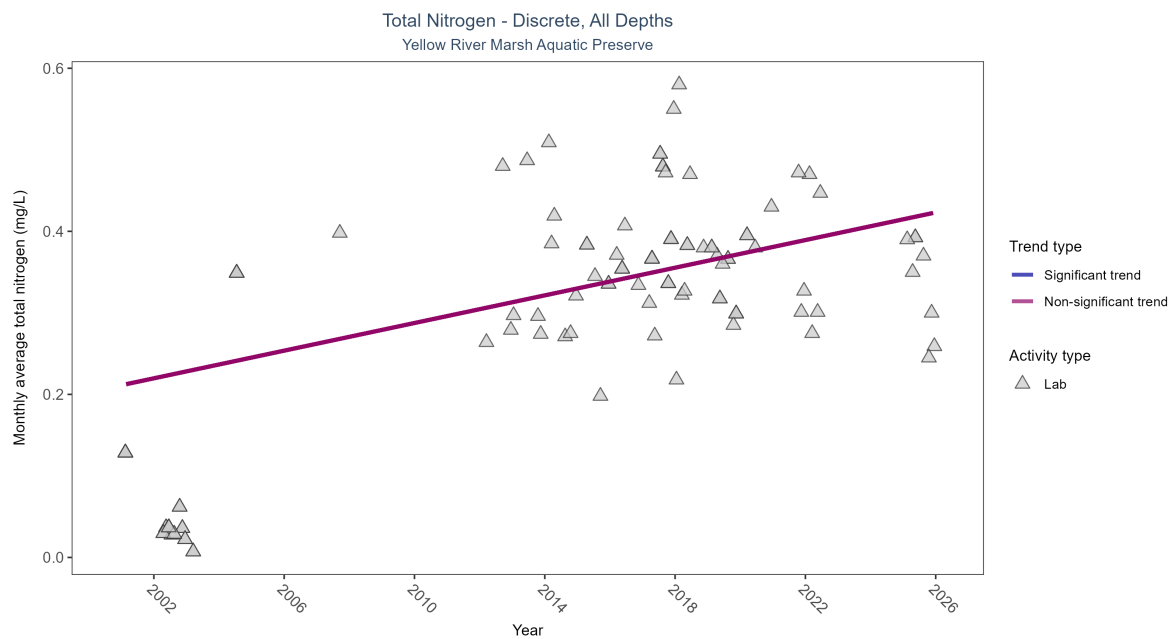


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Seasonal Kendall-Tau Results for - Total Nitrogen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	116	18	2001 - 2025	0.3245	0.18044	0.21123	0.00848	0.0519

Total nitrogen showed no detectable trend between 2001 and 2025.

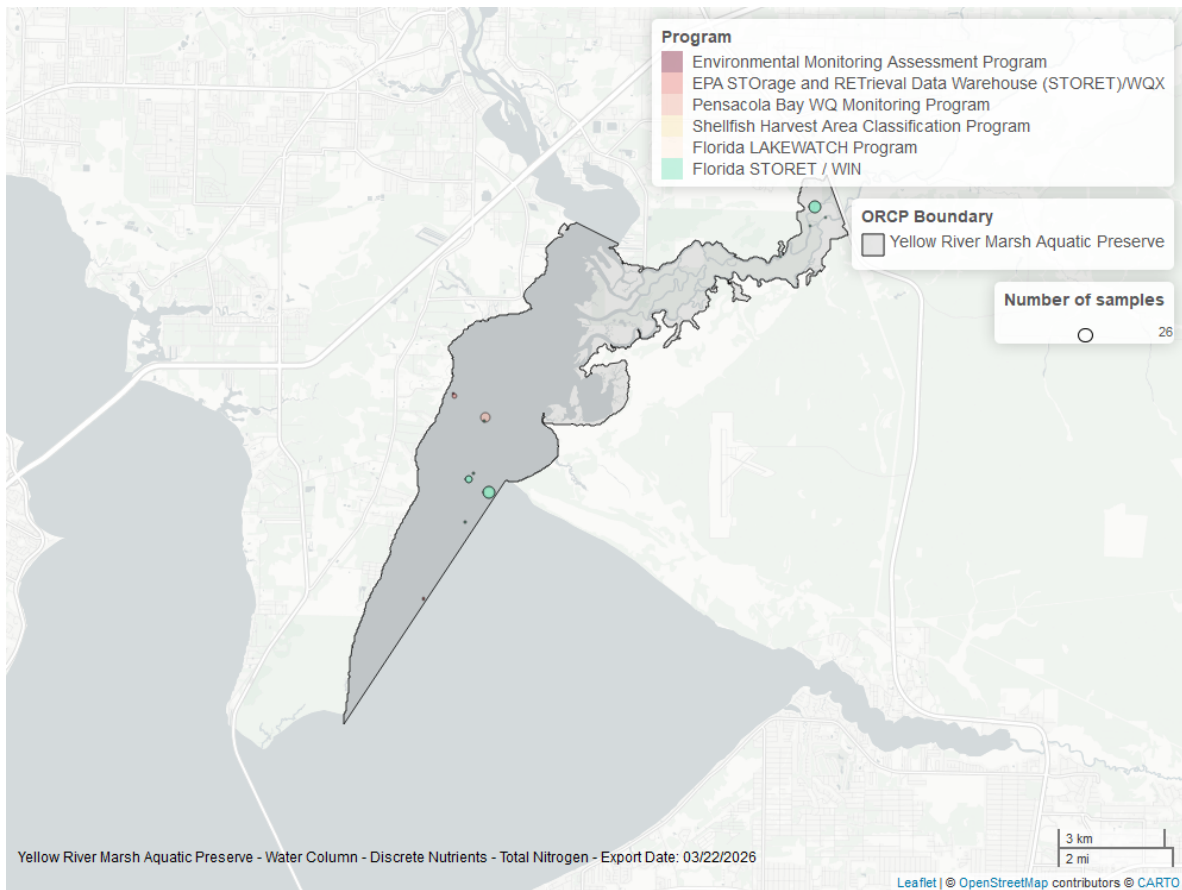


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

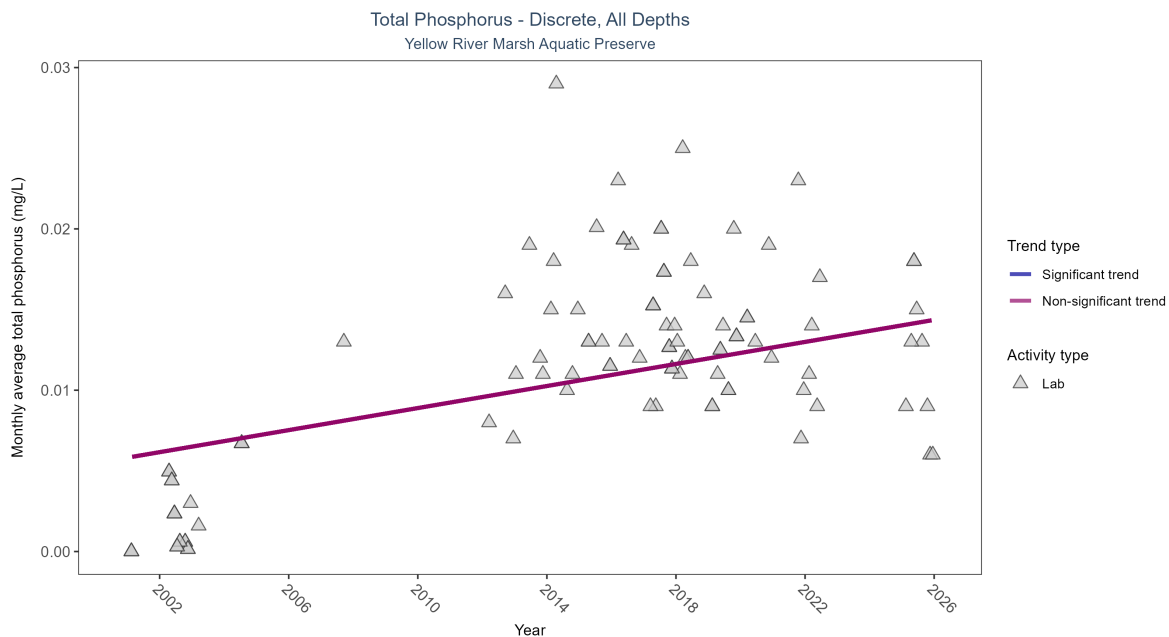


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Seasonal Kendall-Tau Results for - Total Phosphorus

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	112	17	2001 - 2025	0.01185	0.19668	0.00582	0.00034	0.0953

Total phosphorus showed no detectable trend between 2001 and 2025.

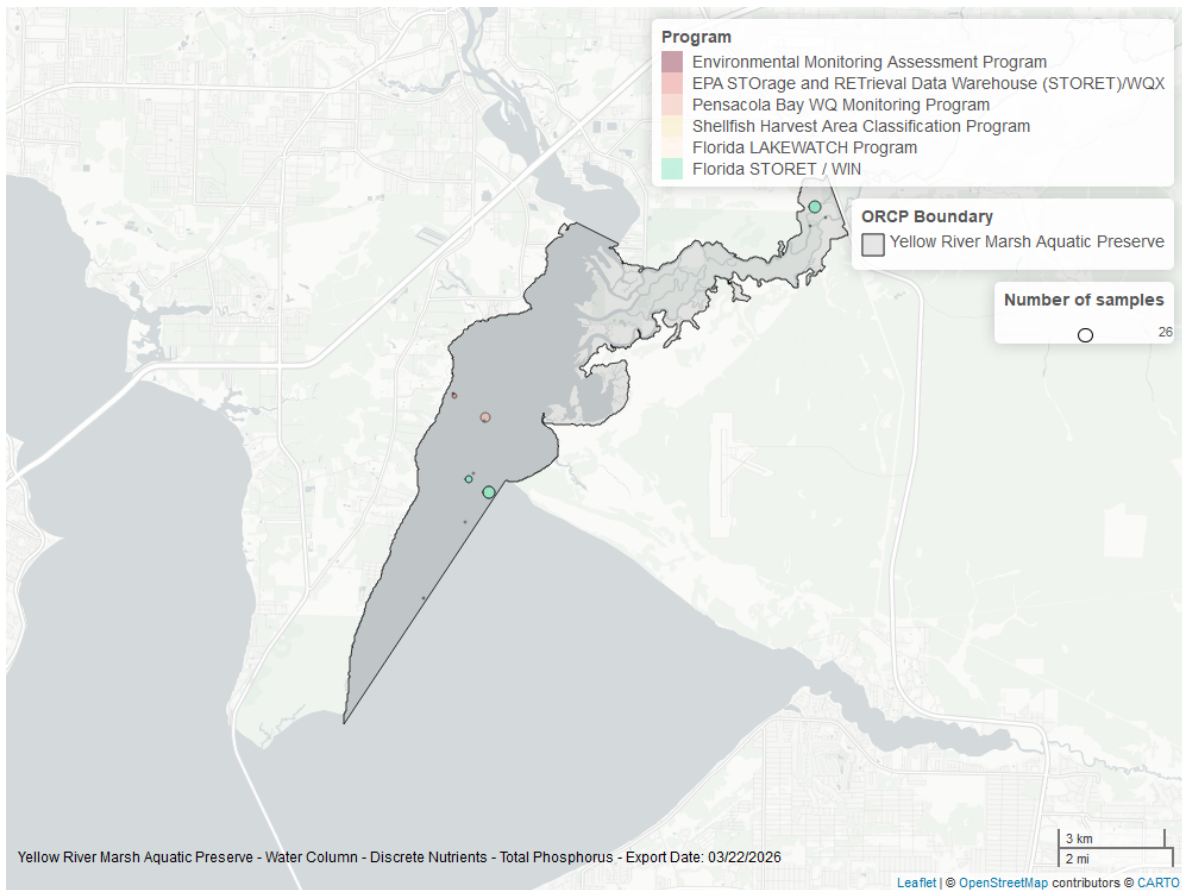


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

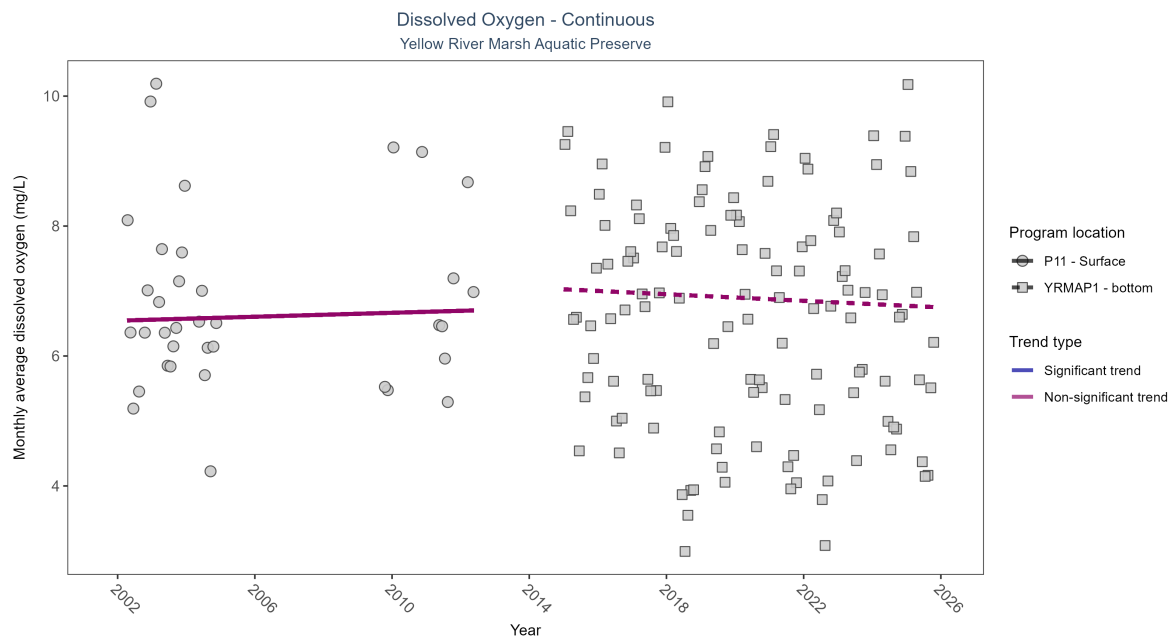


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: Seasonal Kendall-Tau Results - Dissolved Oxygen

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
YRMAP1	No significant trend	319189	11	2015 - 2025	6.80	-0.10	7.03	-0.03	0.1221
P11	No significant trend	131	7	2002 - 2012	6.37	0.04	6.54	0.02	0.4884

No detectable change in monthly average dissolved oxygen was observed at two locations.

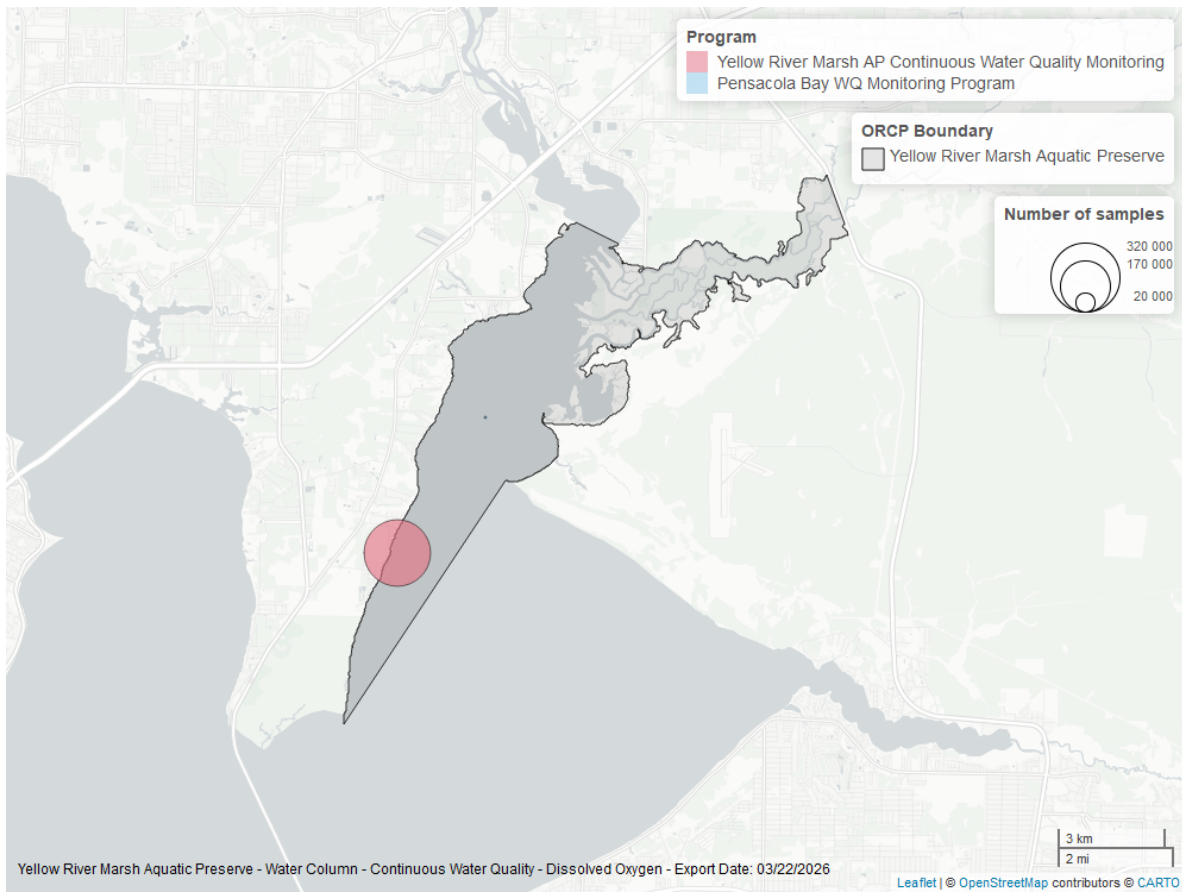


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

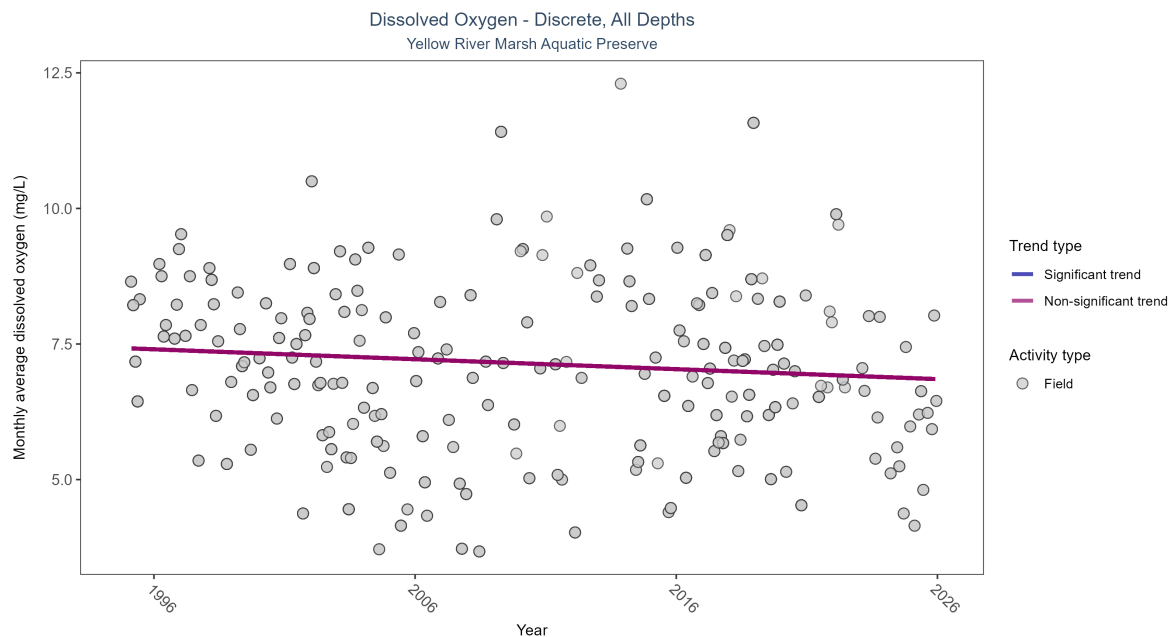


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Seasonal Kendall-Tau Results for - Dissolved Oxygen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No significant trend	1129	31	1995 - 2025	7.18	-0.05916	7.42086	-0.01833	0.0837

Dissolved oxygen showed no detectable trend between 1995 and 2025.

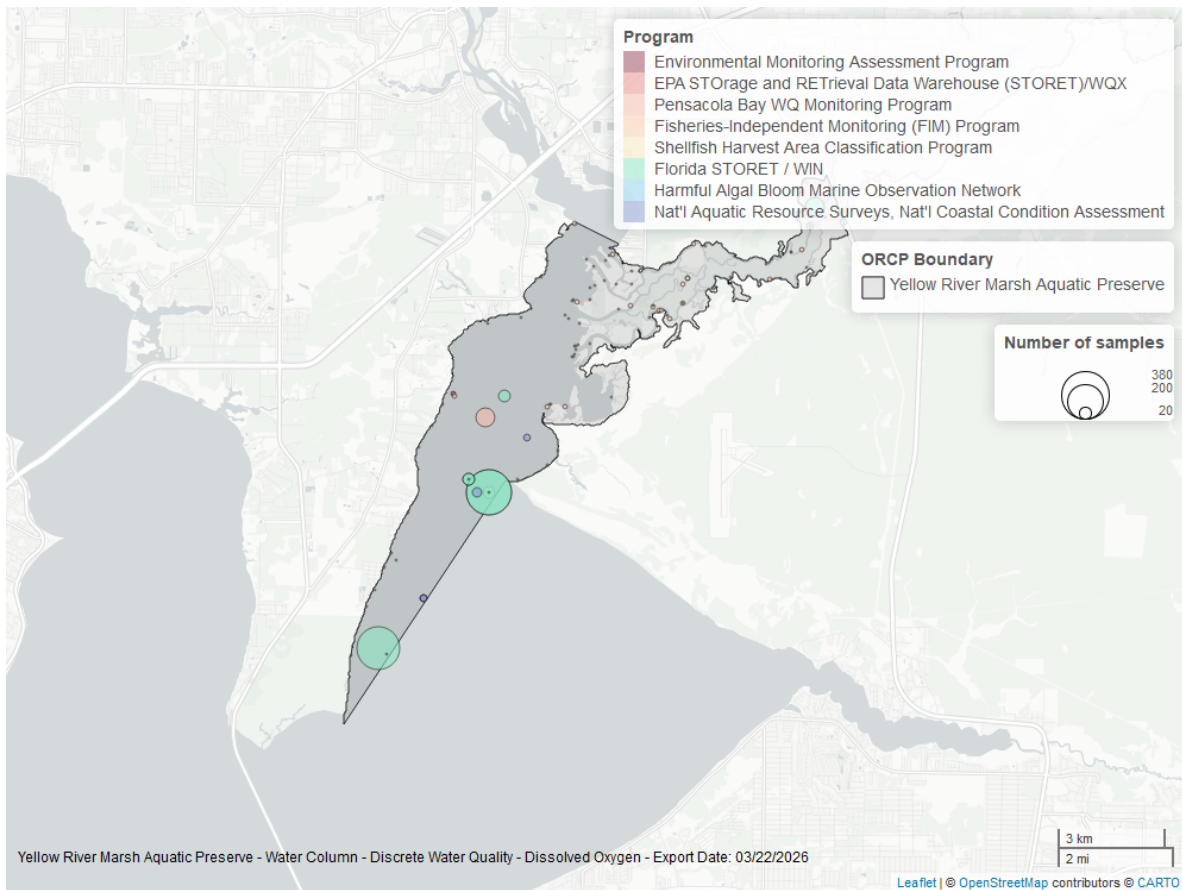


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

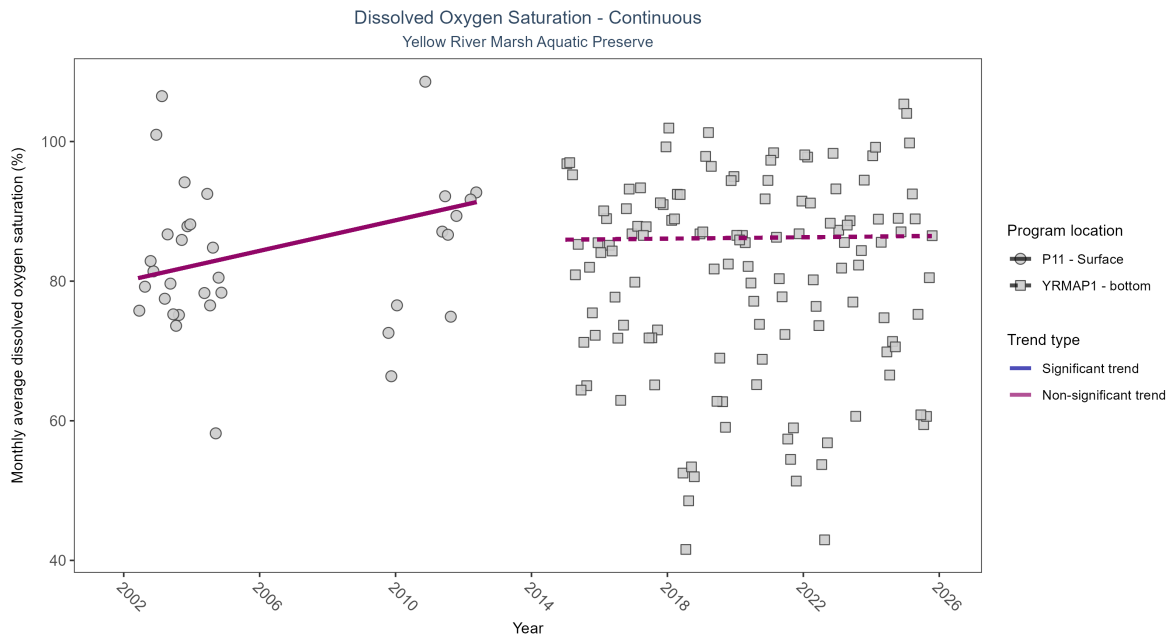


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: Seasonal Kendall-Tau Results - Dissolved Oxygen Saturation

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
YRMAP1	No significant trend	325454	11	2015 - 2025	86.60	0.02	85.94	0.05	0.8323
P11	No significant trend	126	7	2002 - 2012	79.93	0.09	79.97	1.09	0.7119

No detectable change in monthly average dissolved oxygen saturation was observed at two locations.

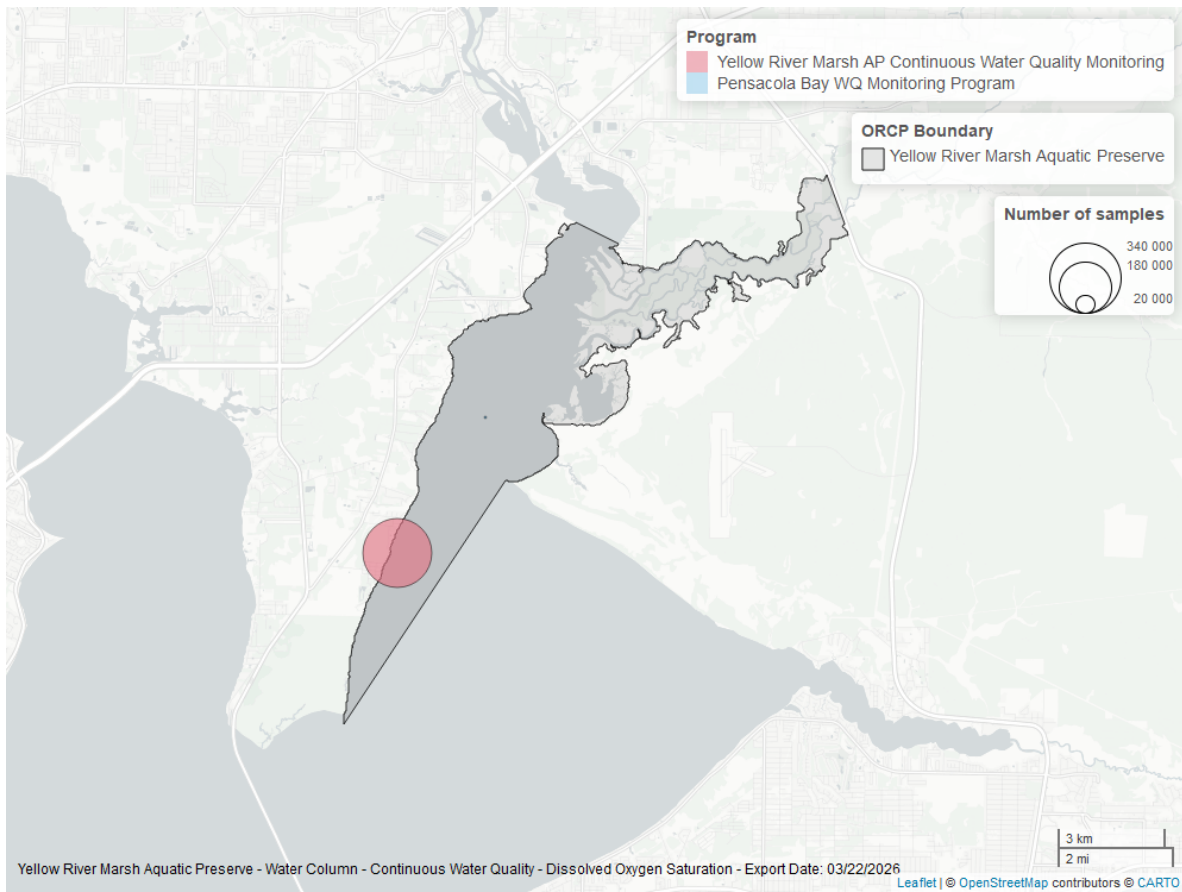


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

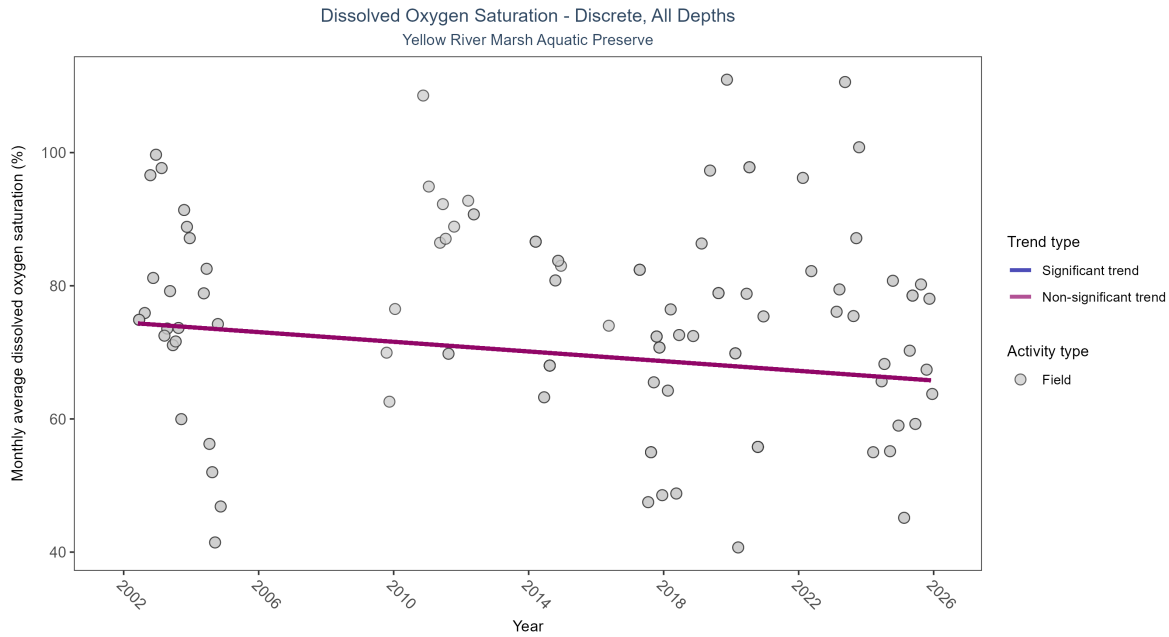


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 6: Seasonal Kendall-Tau Results for - Dissolved Oxygen Saturation

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No significant trend	193	17	2002 - 2025	75.38	-0.10155	74.49593	-0.36364	0.2239

Dissolved oxygen saturation showed no detectable trend between 2002 and 2025.

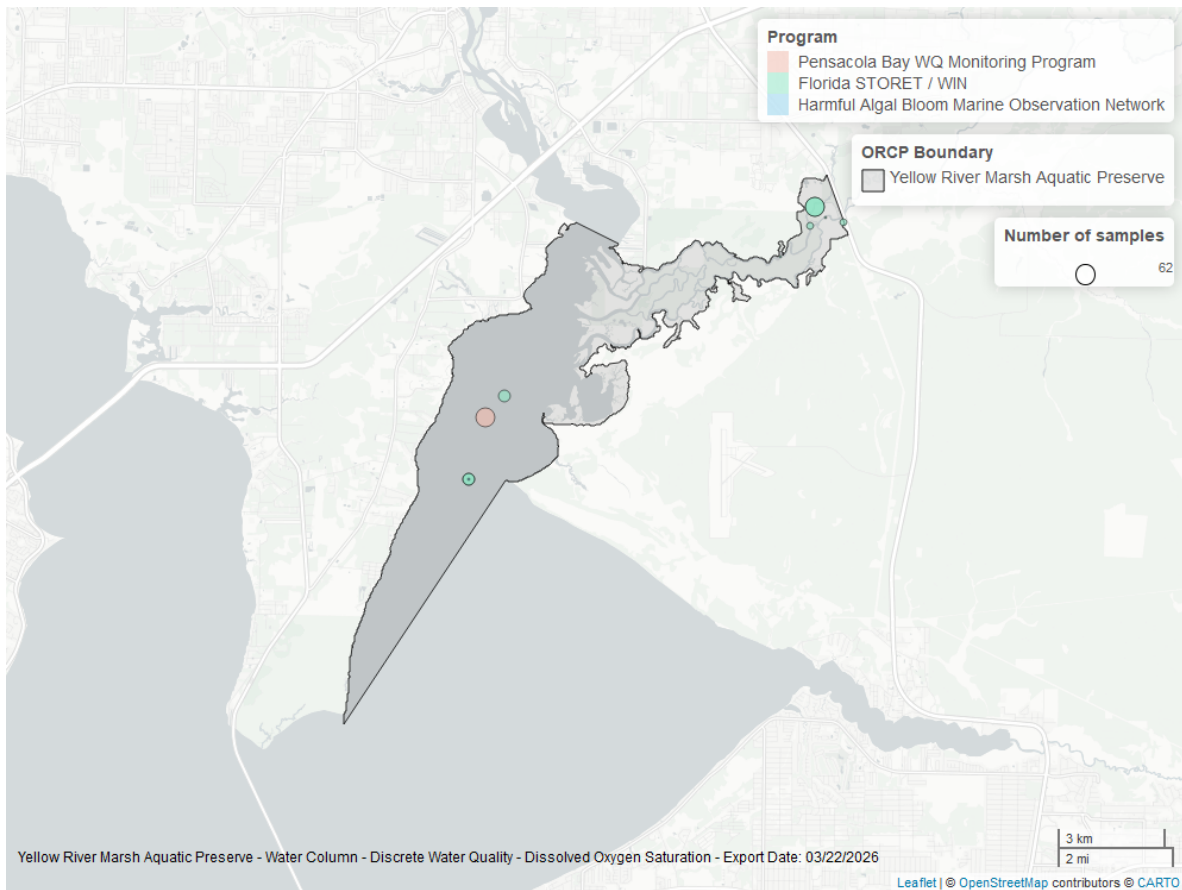


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

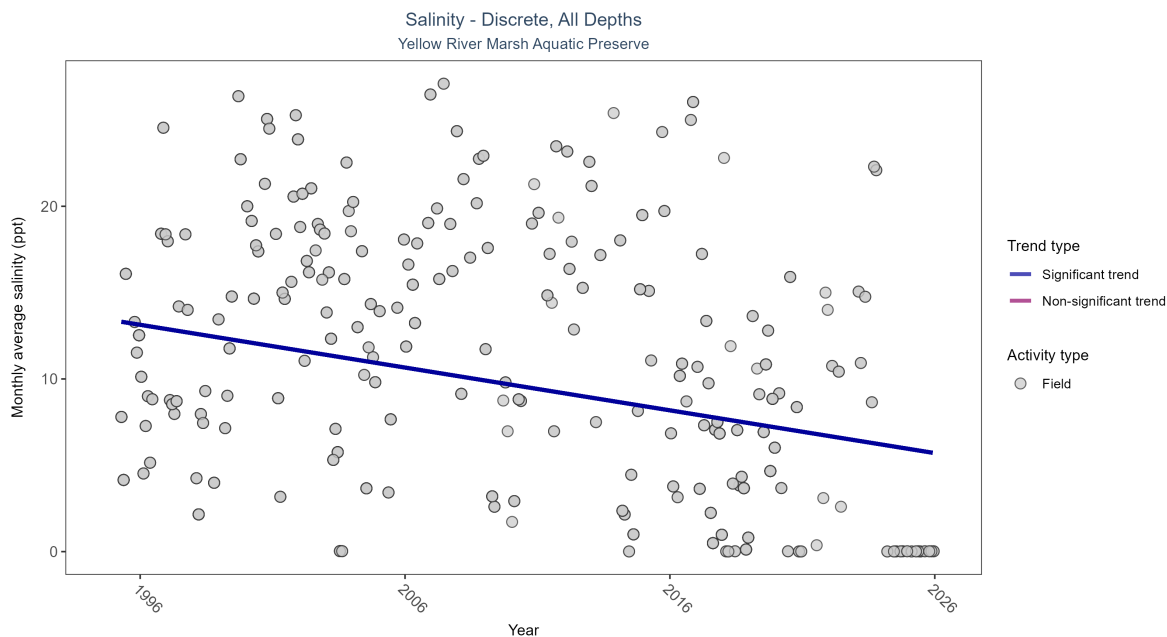


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 7: Seasonal Kendall-Tau Results for - Salinity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Significantly decreasing trend	1260	31	1995 - 2025	10.9	-0.17695	13.38403	-0.2477	0.0001

Monthly average salinity decreased by 0.25 ppt per year.

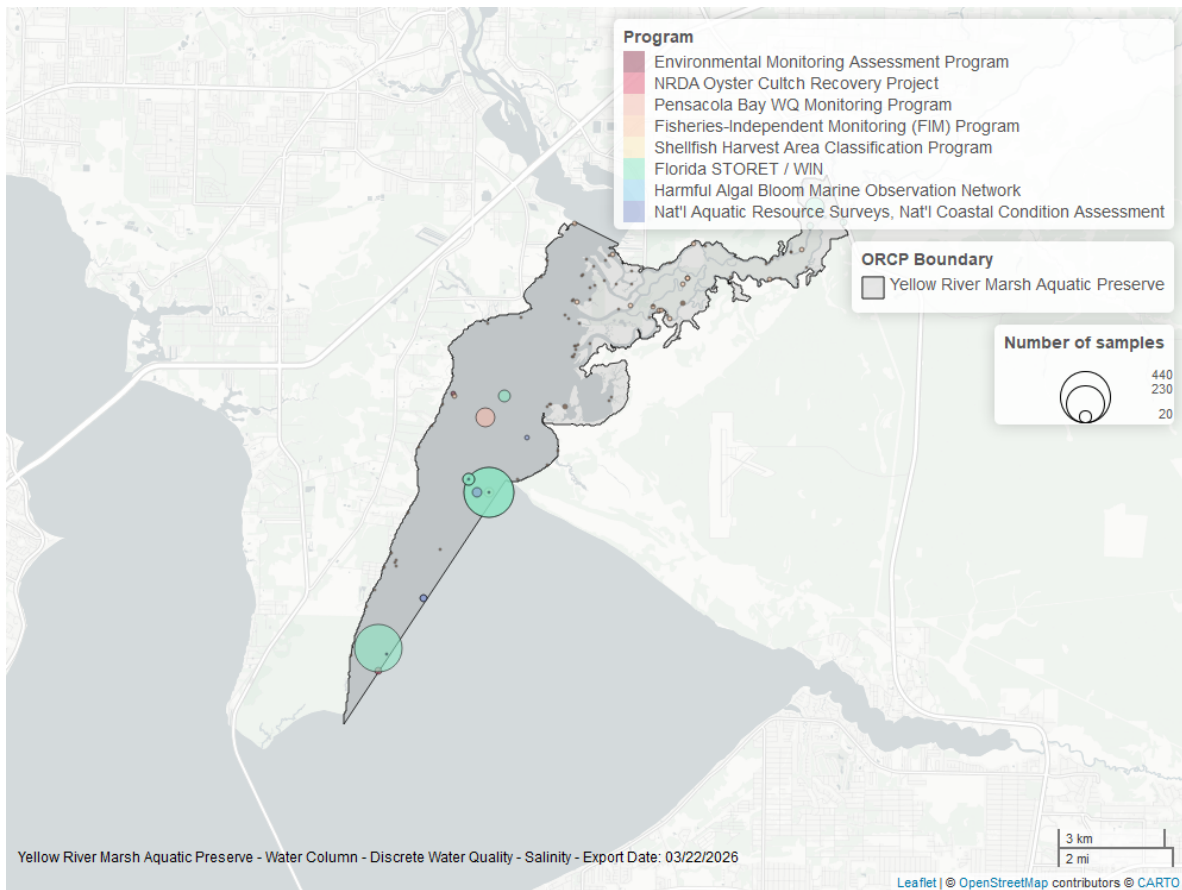


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

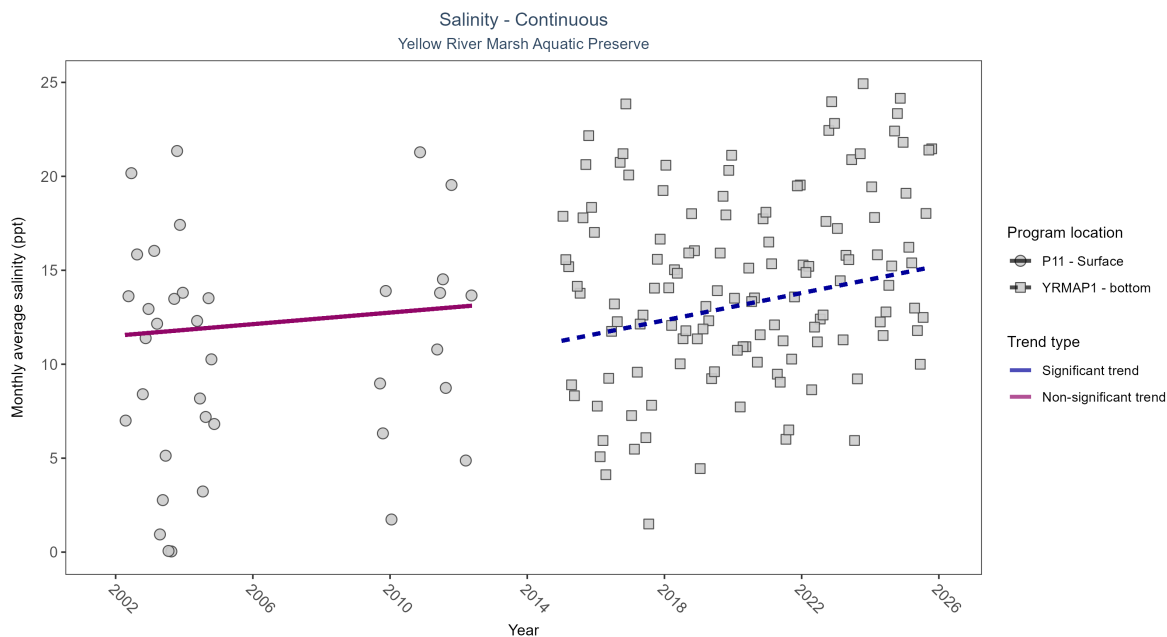


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: Seasonal Kendall-Tau Results - Salinity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
YRMAP1	Significantly increasing trend	331836	11	2015 - 2025	14.50	0.26	11.24	0.36	0.0003
P11	No significant trend	136	7	2002 - 2012	10.05	0.06	11.52	0.15	0.6499

At one program location, monthly average salinity increased by 0.36 ppt per year. No detectable change in monthly average salinity was observed at one location.

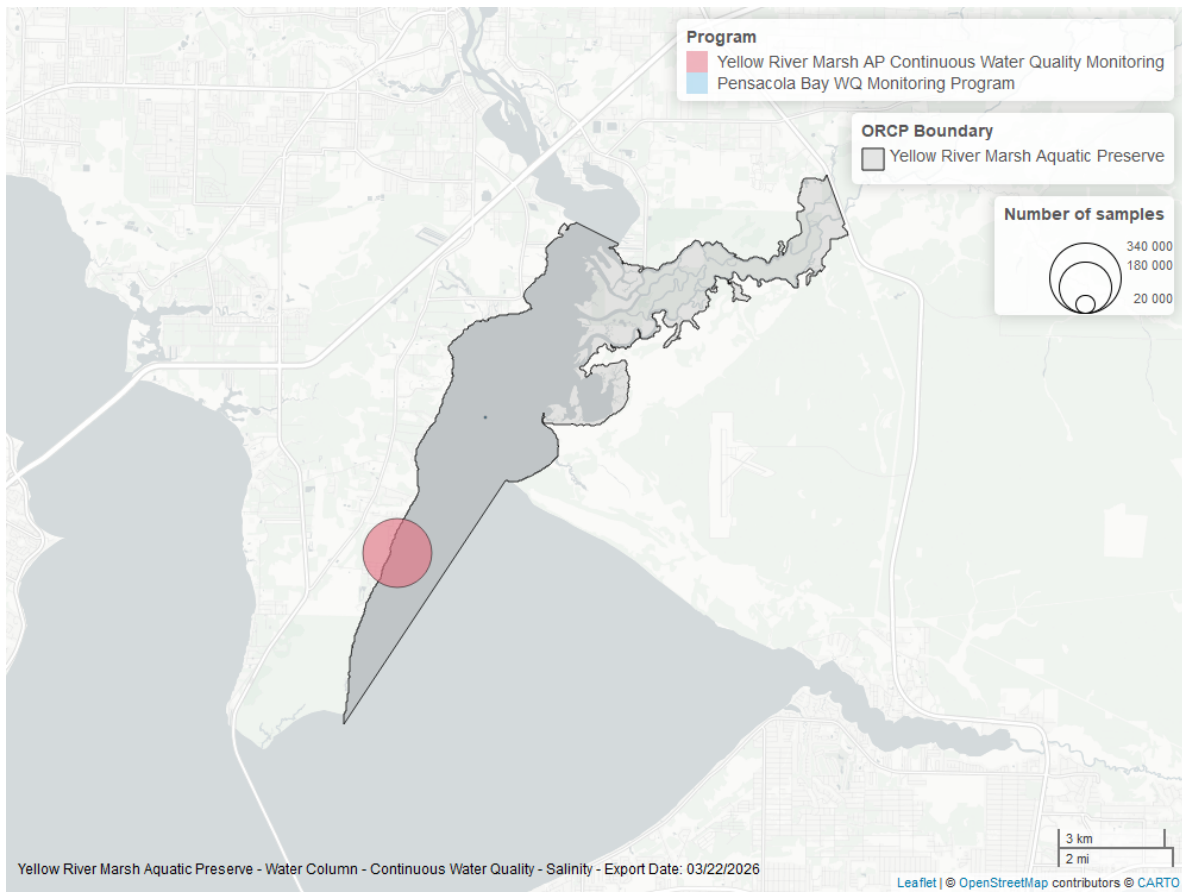


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

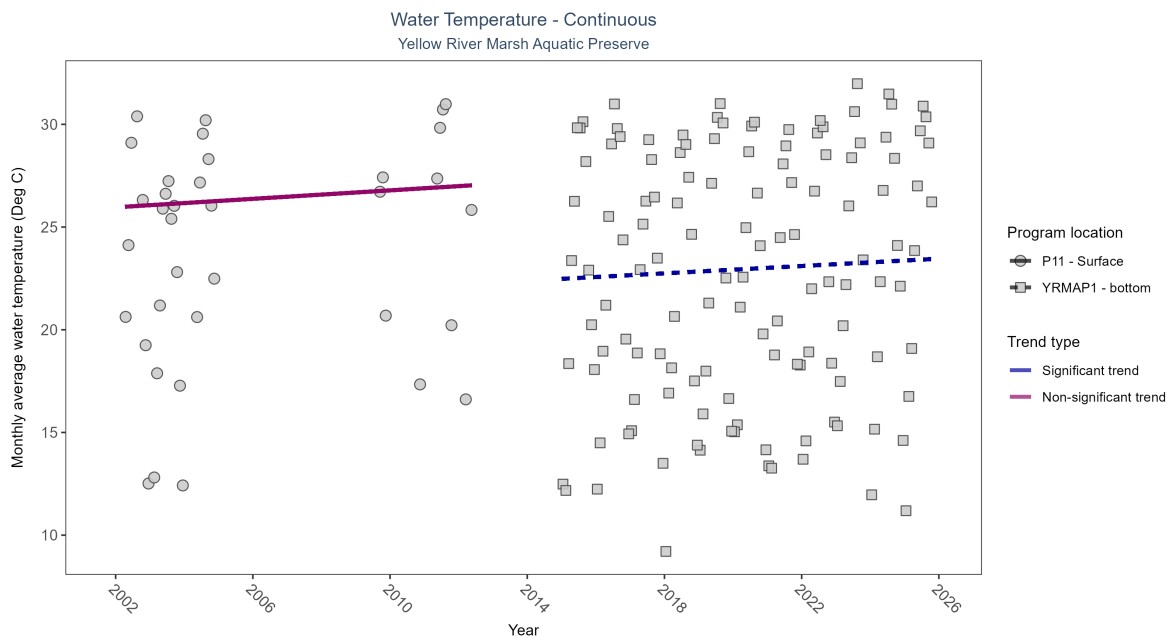


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: Seasonal Kendall-Tau Results - Water Temperature

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
YRMAP1	Significantly increasing trend	362580	11	2015 - 2025	23.30	0.16	22.48	0.09	0.0208
P11	No significant trend	136	7	2002 - 2012	26.22	0.13	25.96	0.10	0.496

At one program location, monthly average water temperature increased by 0.09°C per year. No detectable change in monthly average water temperature was observed at one location.

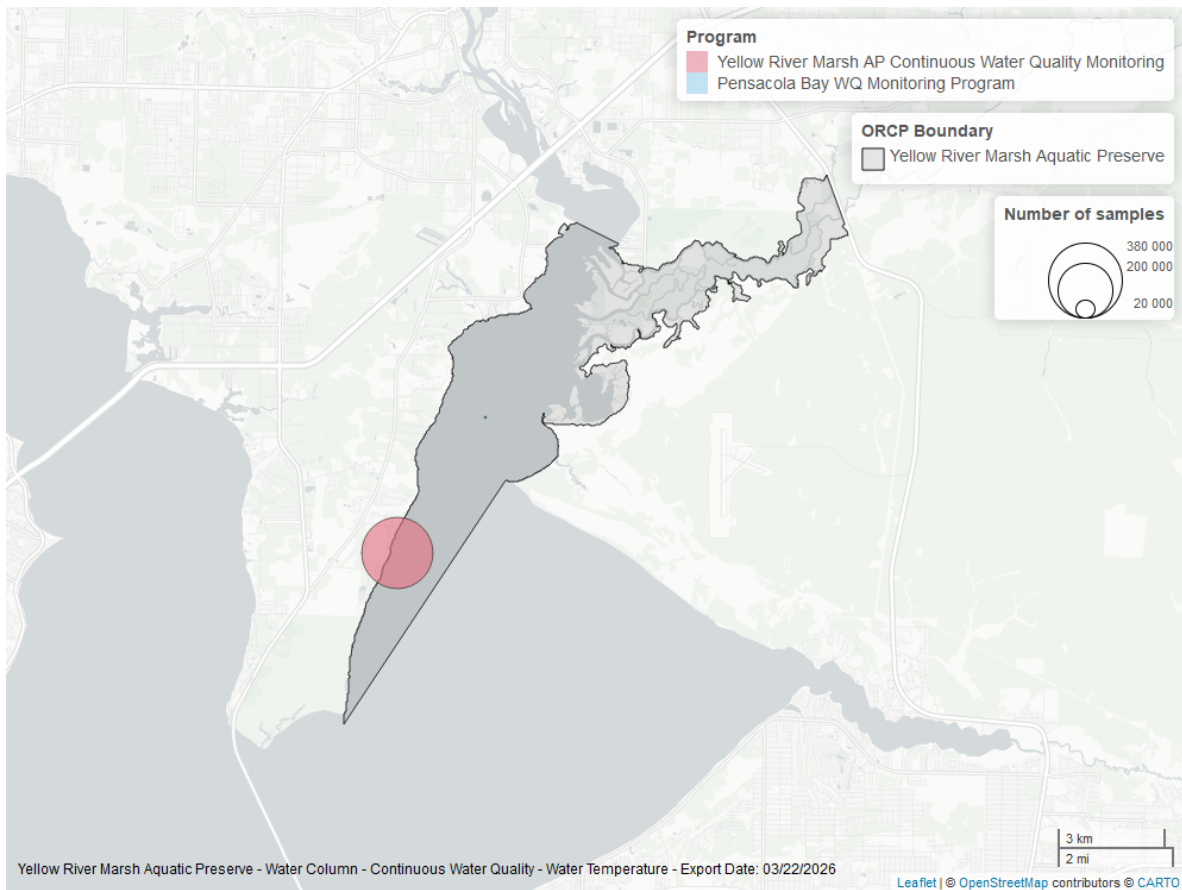


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

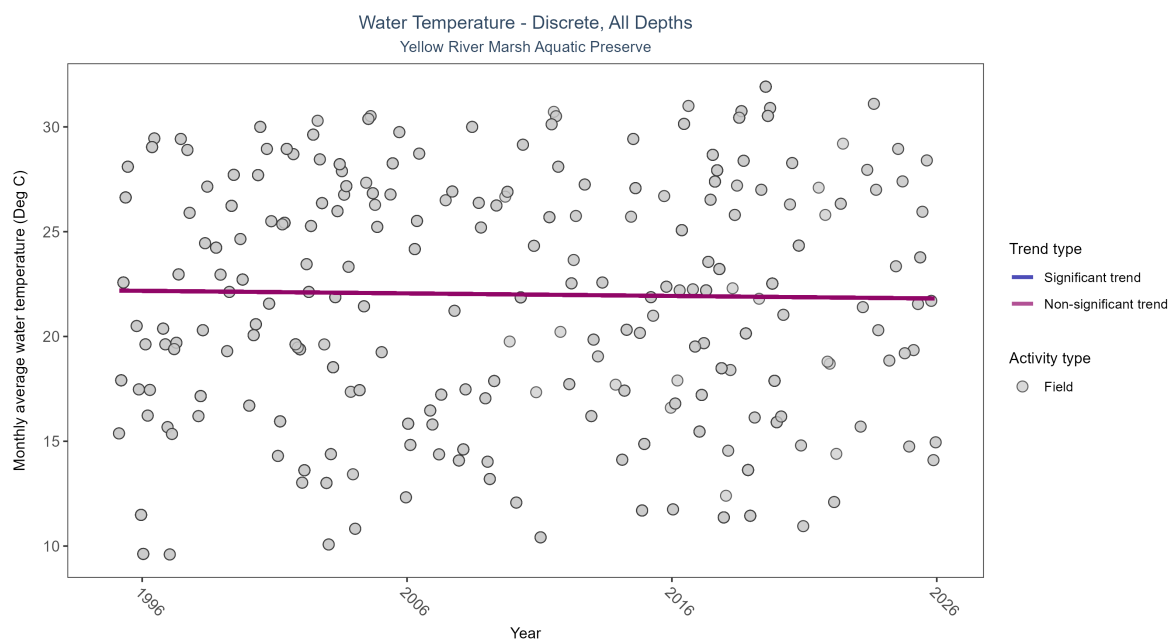


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 10: Seasonal Kendall-Tau Results for - Water Temperature

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No significant trend	1255	31	1995 - 2025	22	-0.02481	22.19026	-0.01221	0.4769

Water temperature showed no detectable trend between 1995 and 2025.

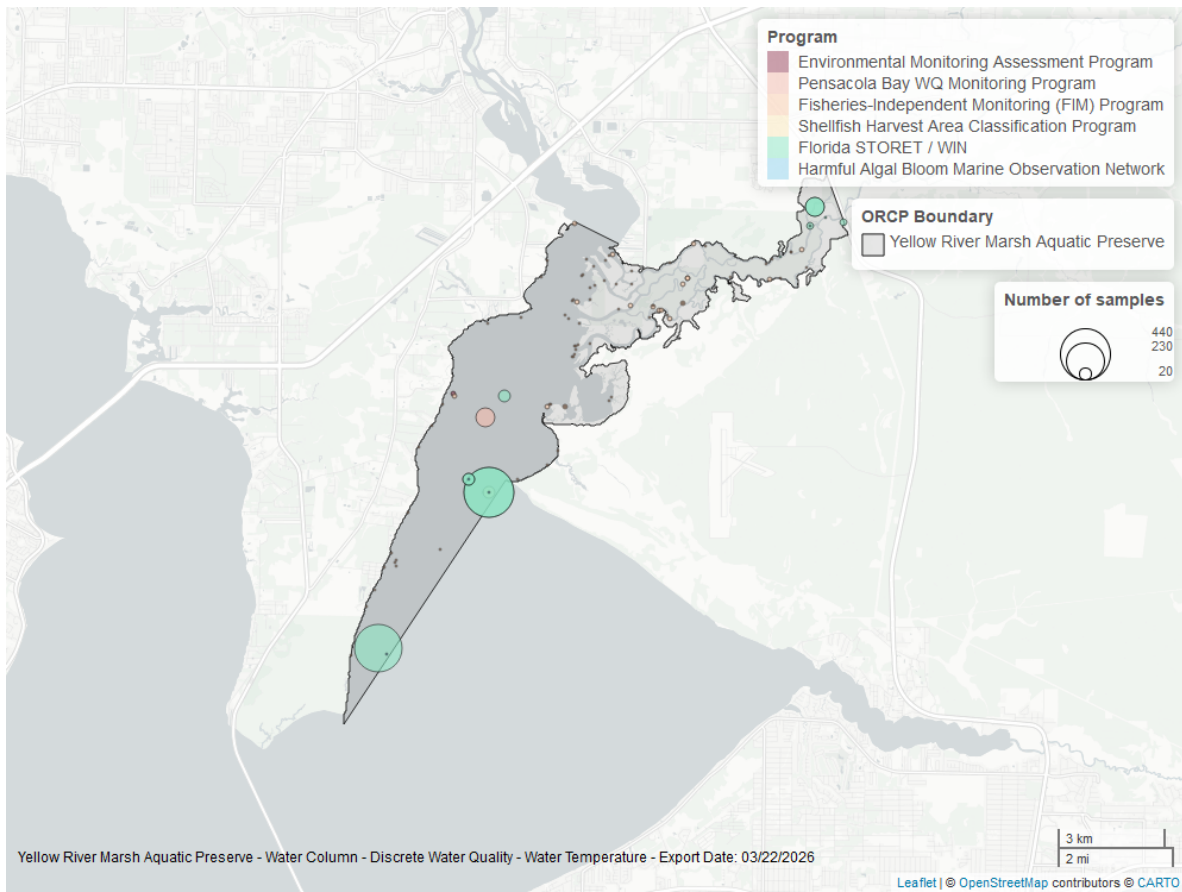


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

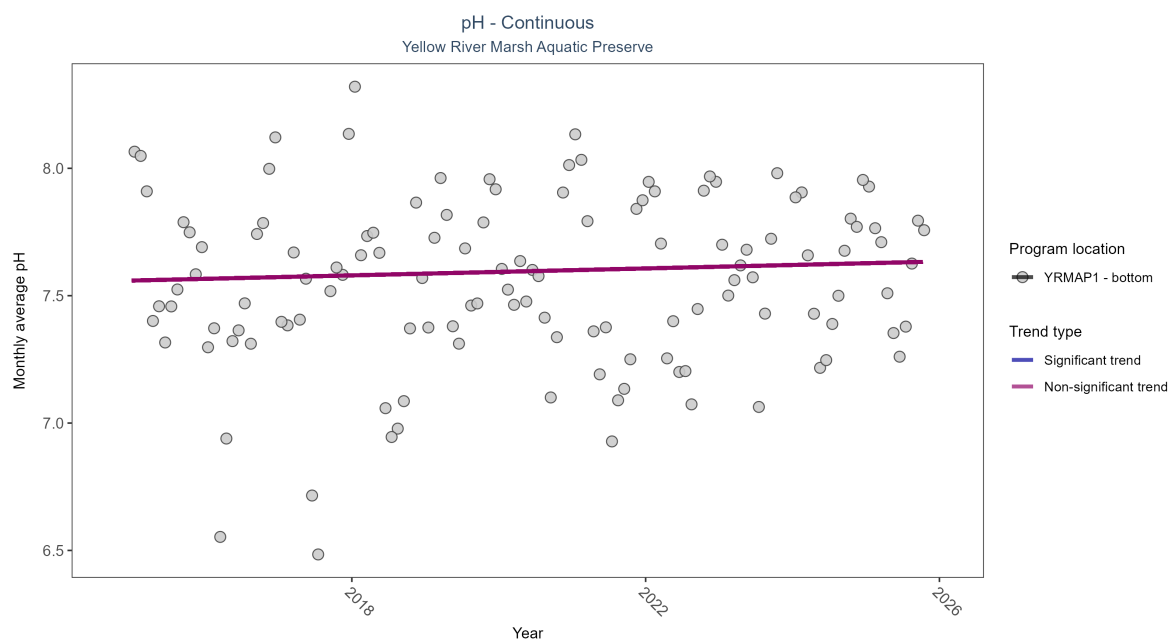


Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: Seasonal Kendall-Tau Results - pH

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
YRMAP1	No significant trend	331267	11	2015 - 2025	7.6	0.07	7.56	0.01	0.333

No detectable change in monthly average pH was observed at one location.

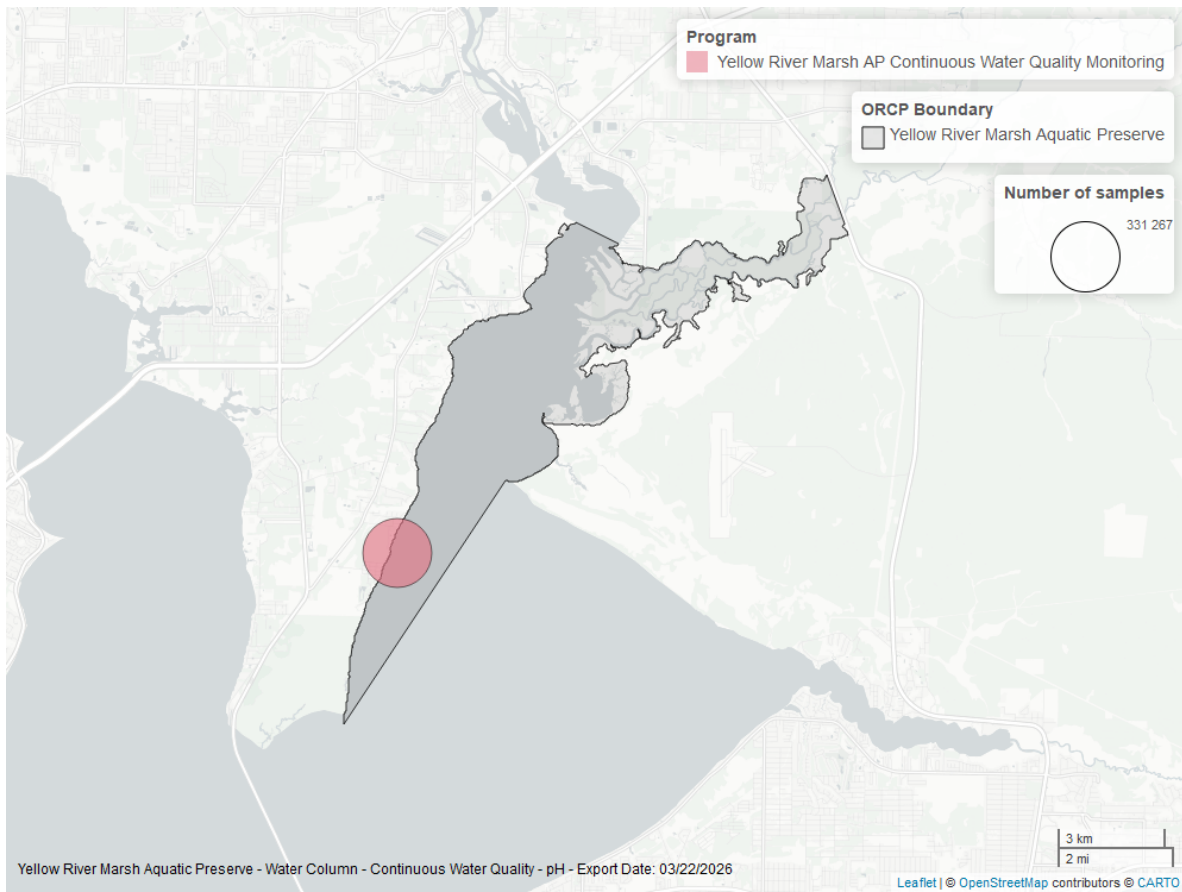


Figure 22: Map showing location of pH continuous water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

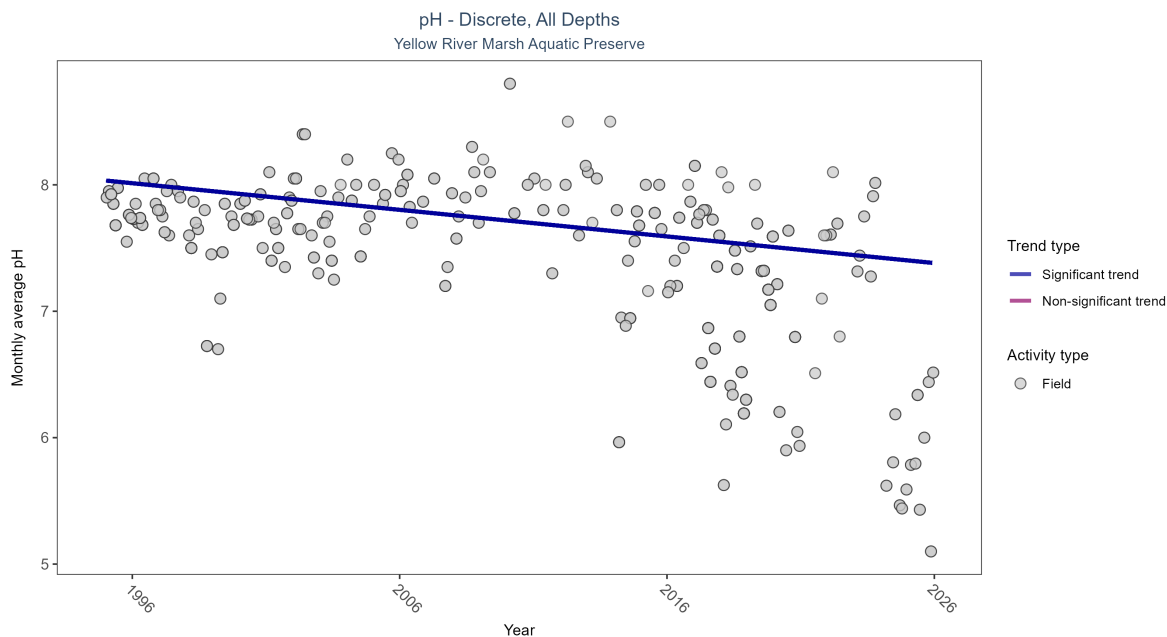


Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Results for - pH

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly decreasing trend	761	31	1995 - 2025	7.6	-0.23417	8.03381	-0.02105	0

Monthly average pH decreased by 0.02 pH units per year.

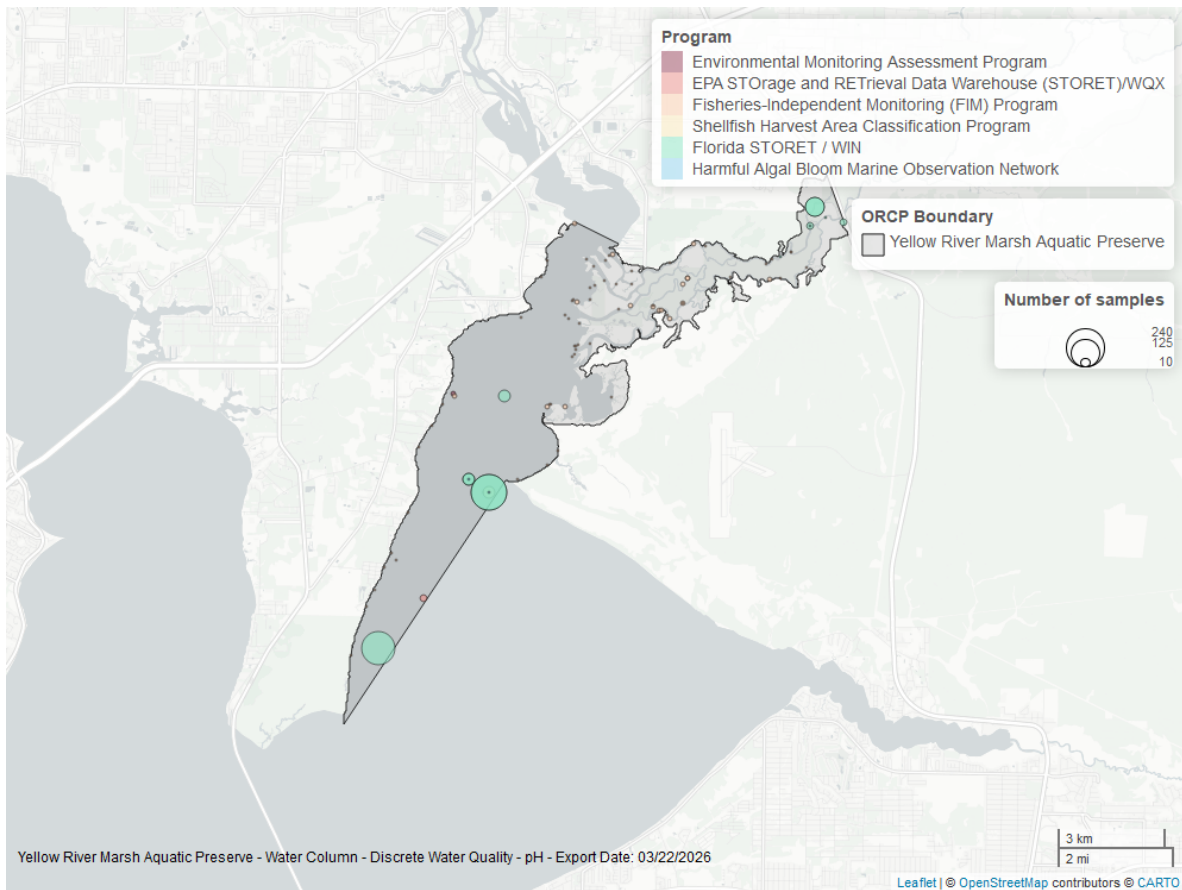


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

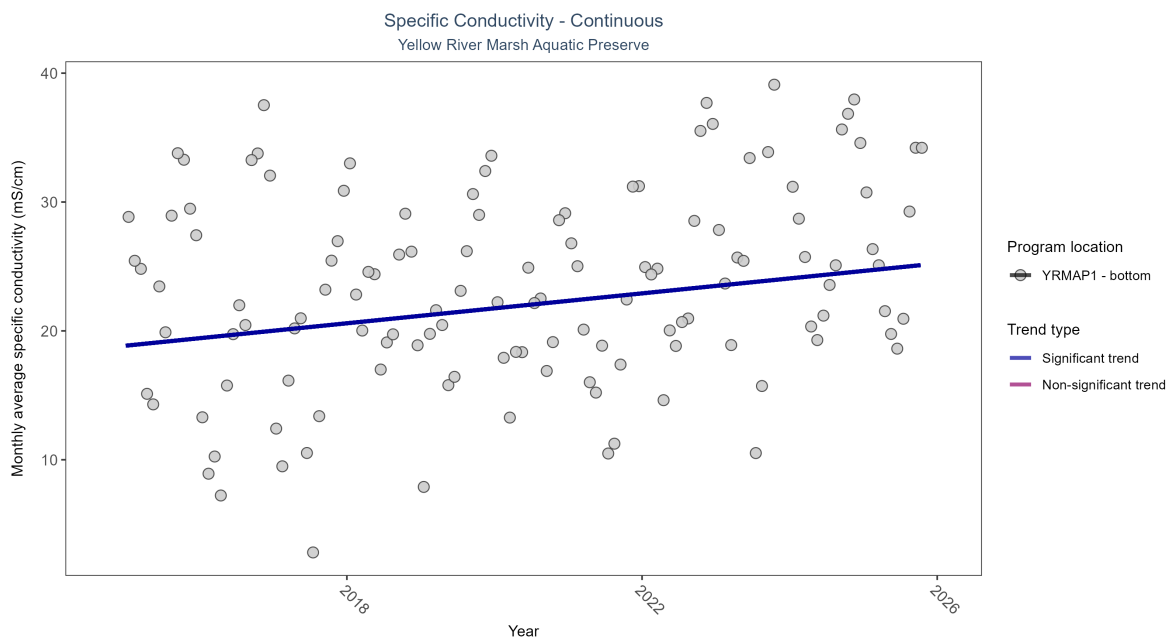


Table 13: Seasonal Kendall-Tau Results - Specific Conductivity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
YRMAP1	Significantly increasing trend	336809	11	2015 - 2025	23.93	0.28	18.85	0.58	0.0001

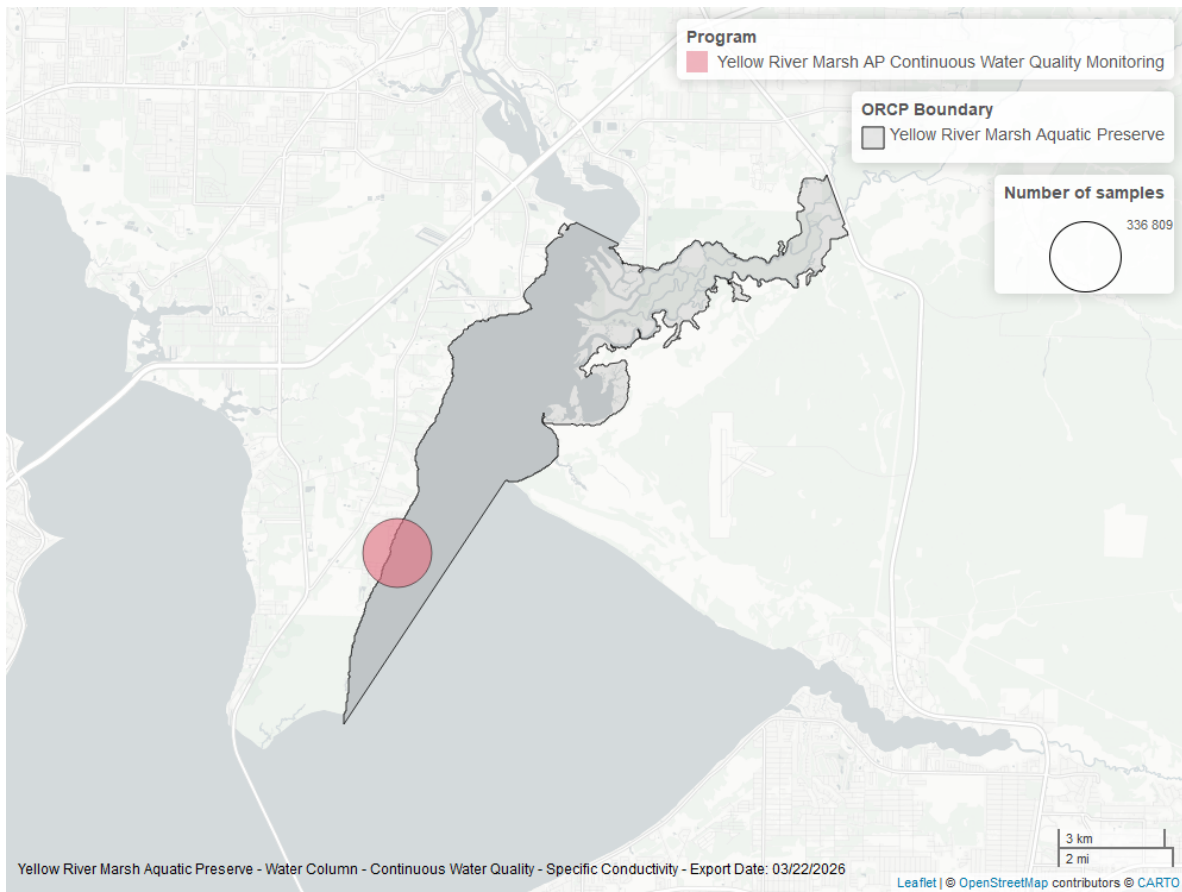


Figure 25: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

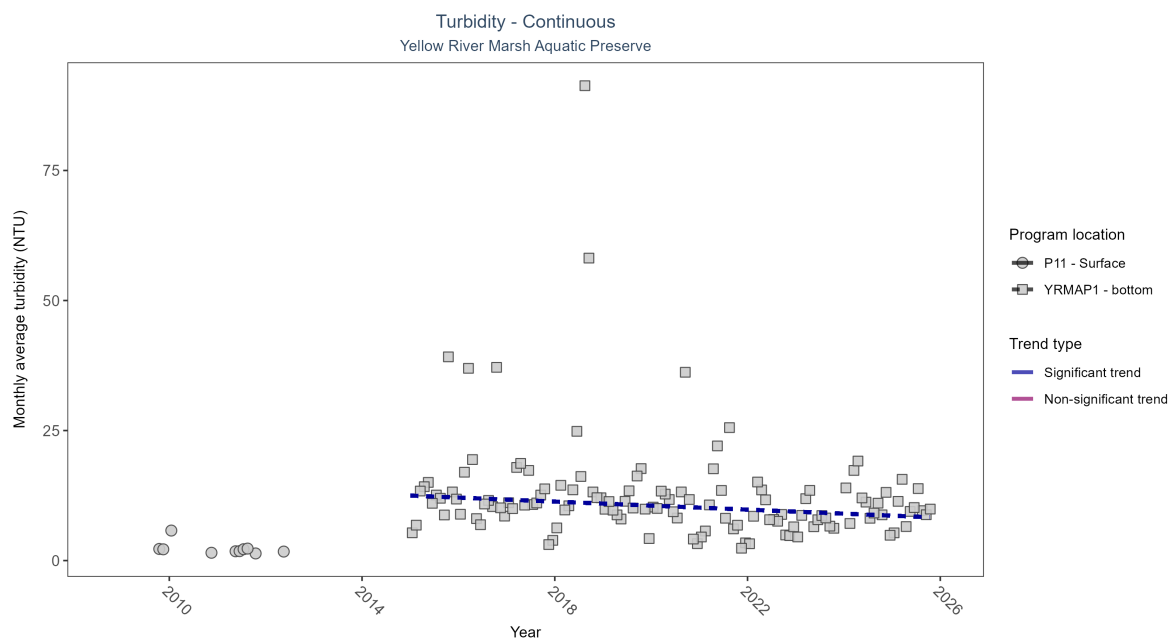


Figure 26: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 14: Seasonal Kendall-Tau Results - Turbidity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
YRMAP1	Significantly decreasing trend	345284	11	2015 - 2025	6	-0.22	12.5	-0.39	0.0016
P11	Insufficient data to calculate trend	37	4	2009 - 2012	2	-	-	-	-

At one program location, monthly average turbidity decreased by 0.39 NTU per year. There was insufficient data to fit a model for one location.

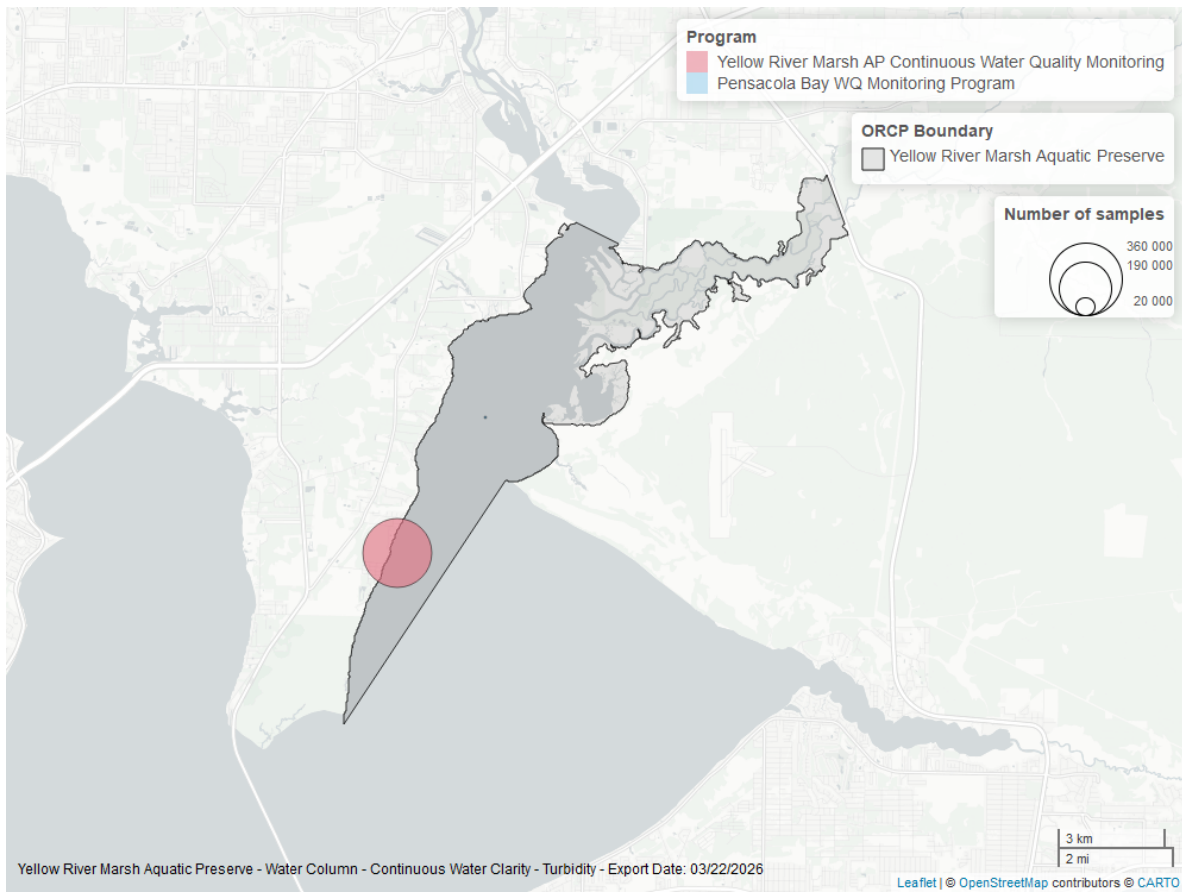


Figure 27: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

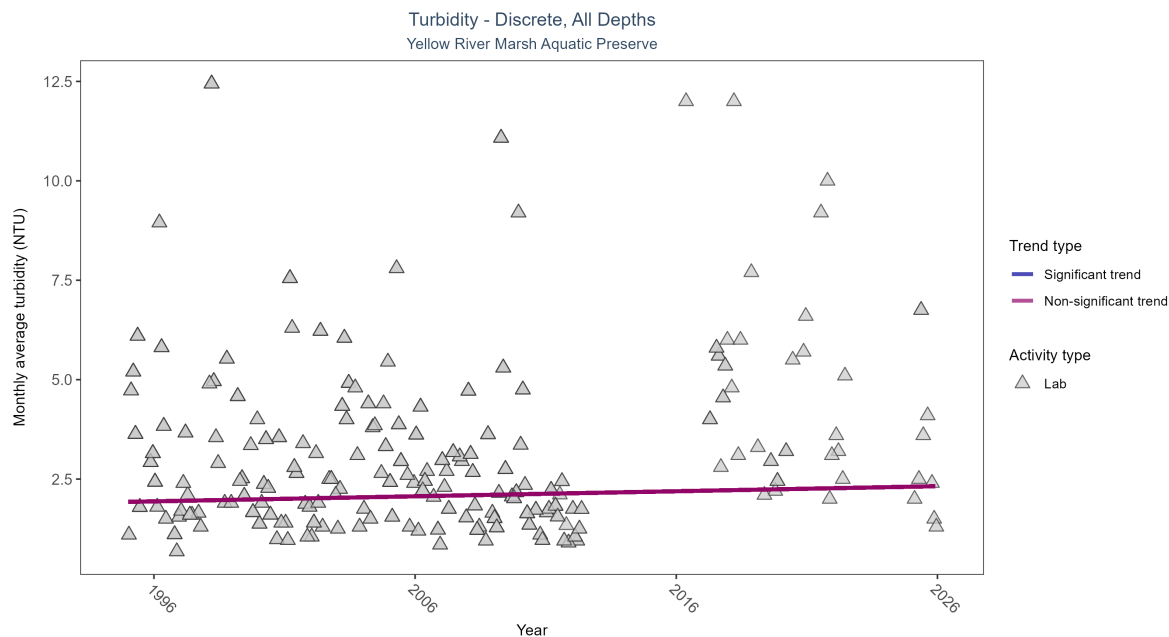


Figure 28: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 15: Seasonal Kendall-Tau Results for - Turbidity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	598	26	1995 - 2025	2.485	0.07412	1.92699	0.0127	0.4436

Turbidity showed no detectable trend between 1995 and 2025.

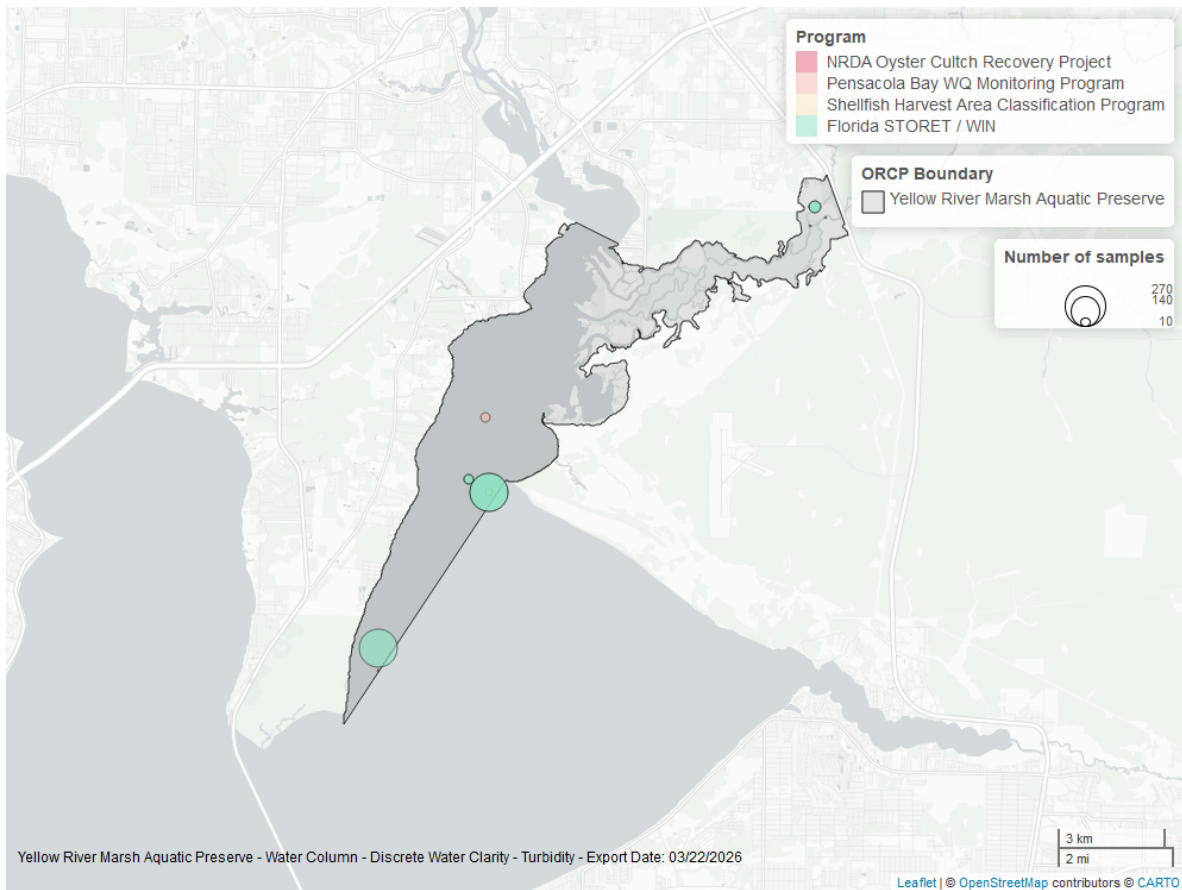


Figure 29: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

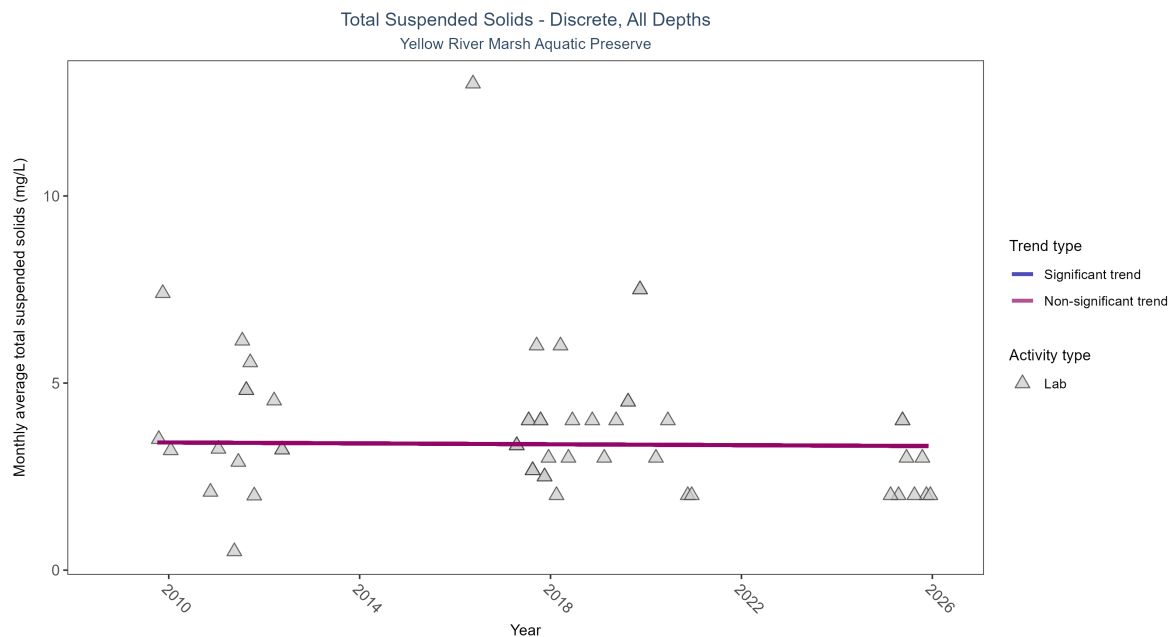


Figure 30: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 16: Seasonal Kendall-Tau Results for - Total Suspended Solids

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	54	10	2009 - 2025	3.17	-0.10952	3.41786	-0.006	0.6315

Total suspended solids showed no detectable trend between 2009 and 2025.

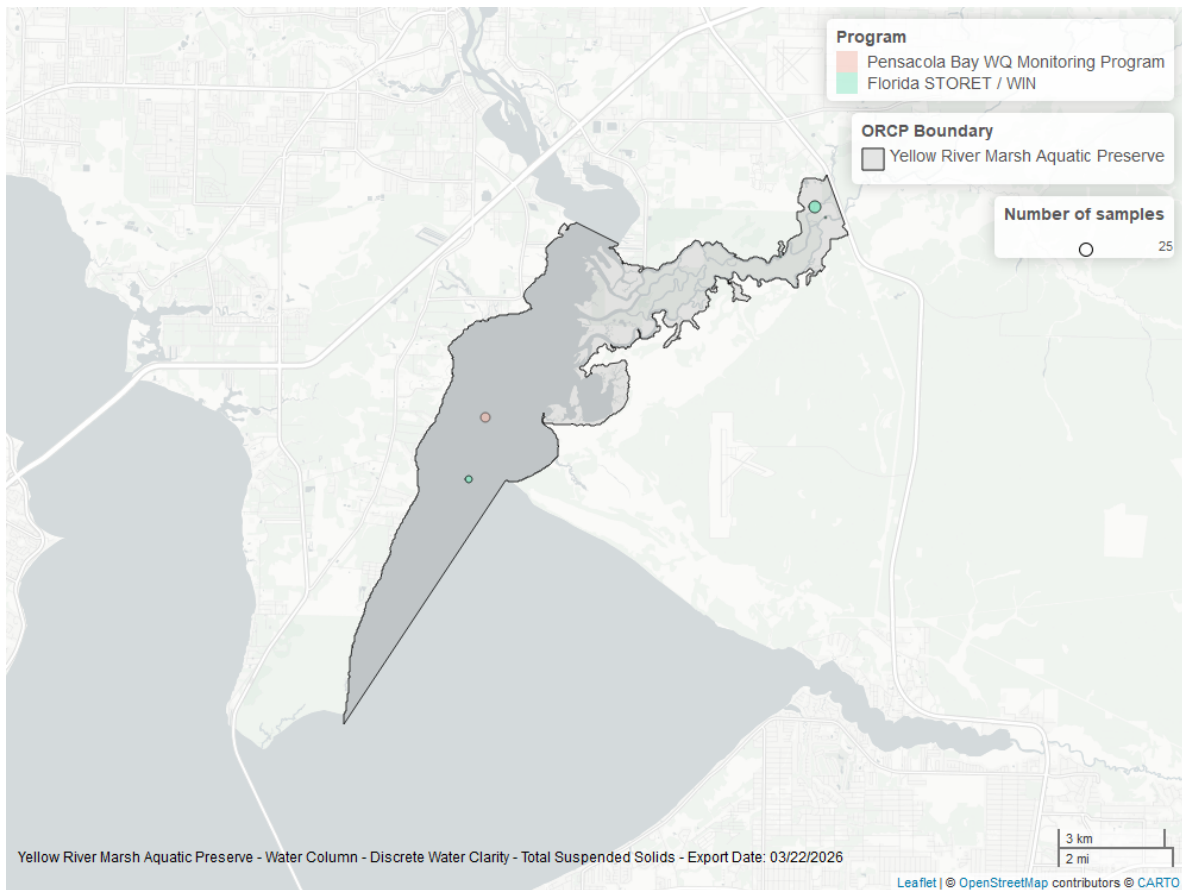


Figure 31: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

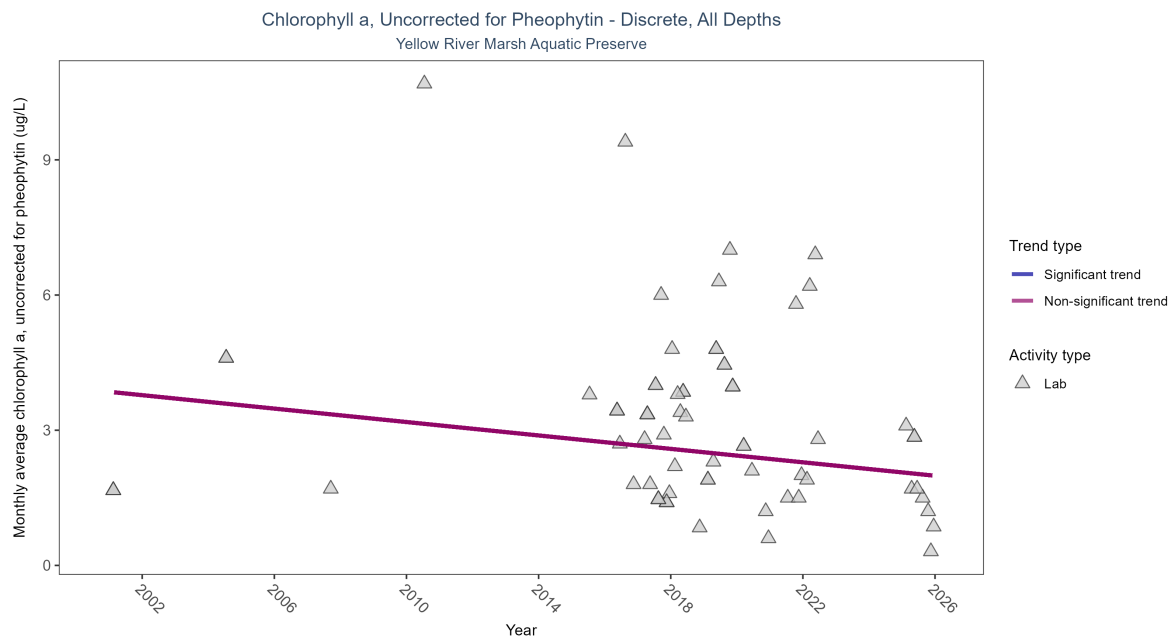


Figure 32: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: Seasonal Kendall-Tau Results for - Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	73	13	2001 - 2025	2.7	-0.09477	3.8525	-0.0745	0.368

Chlorophyll a, uncorrected for pheophytin, showed no detectable trend between 2001 and 2025.

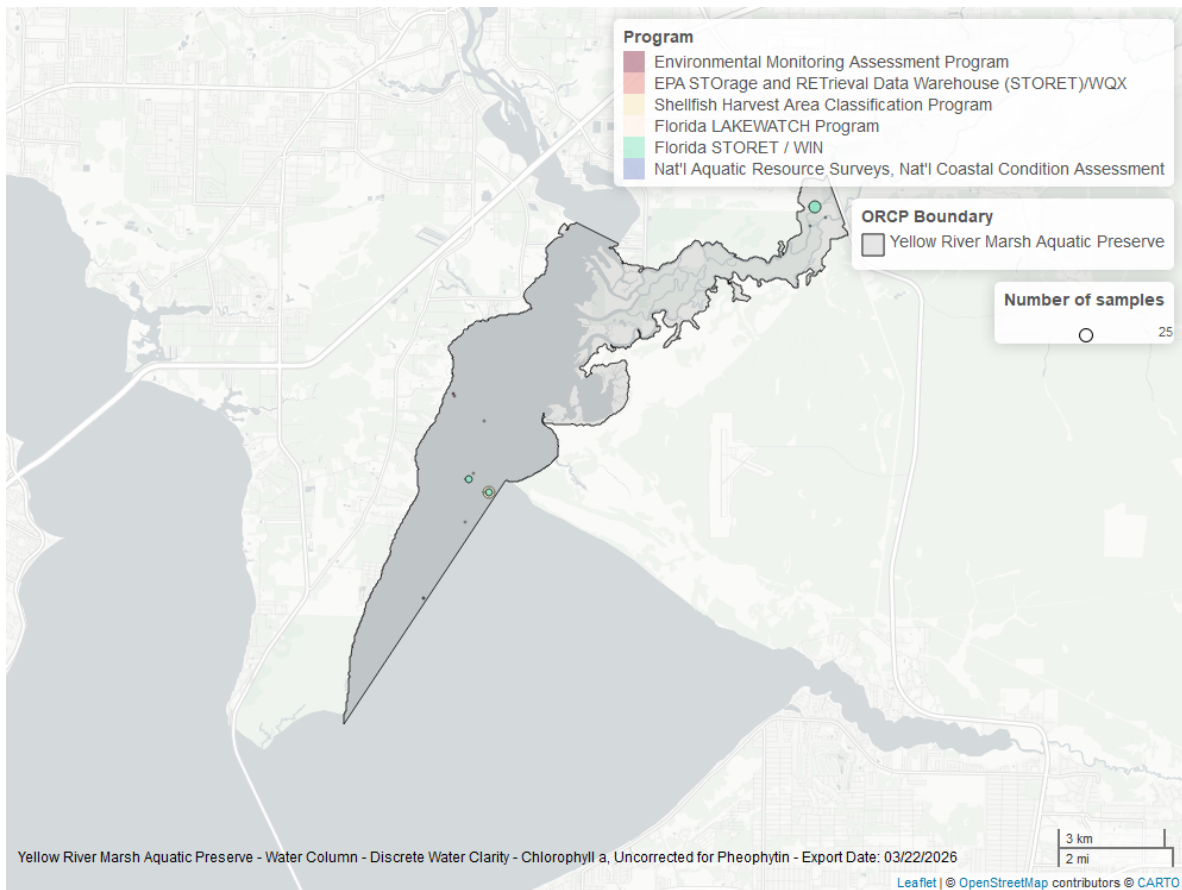


Figure 33: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

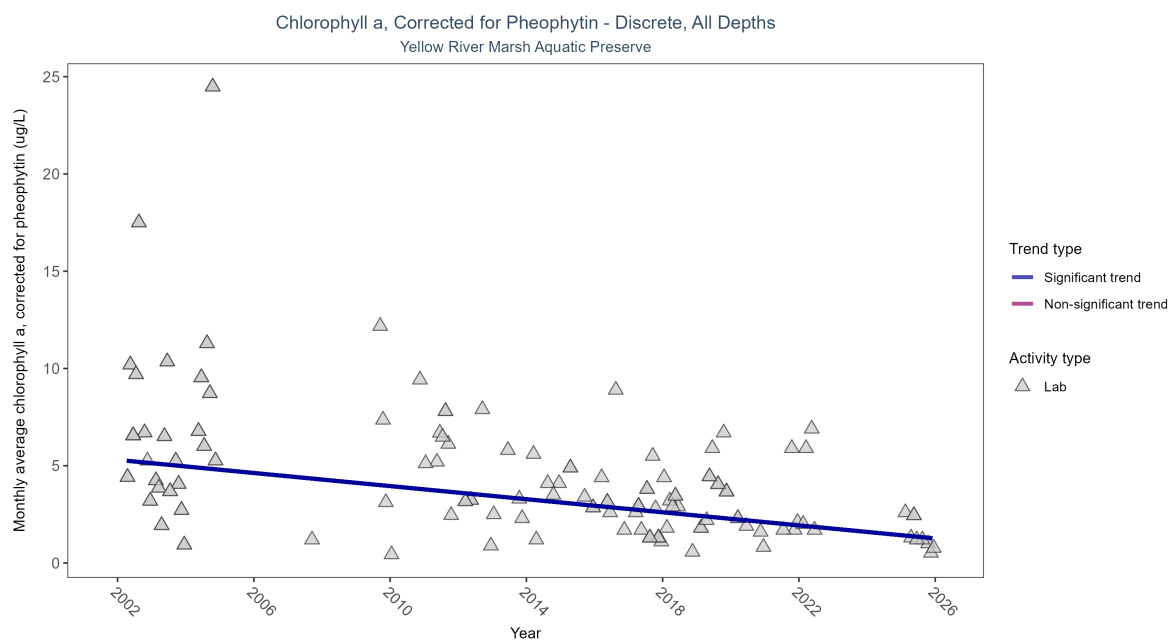


Figure 34: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 18: Seasonal Kendall-Tau Results for - Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly decreasing trend	151	19	2002 - 2025	3.49	-0.36975	5.30057	-0.16833	0

Monthly average chlorophyll a, corrected for pheophytin, decreased by 0.17 $\mu\text{g/L}$ per year, indicating an increase in water clarity.

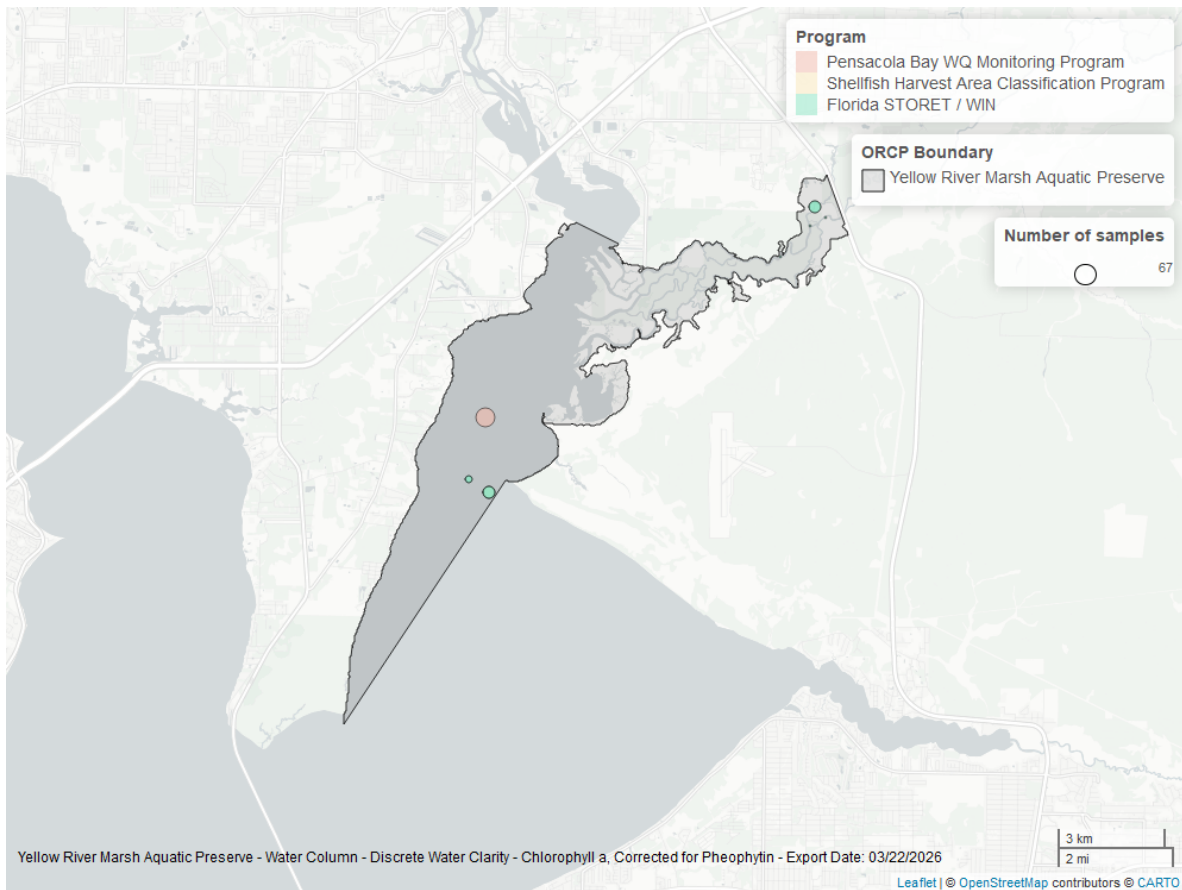


Figure 35: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

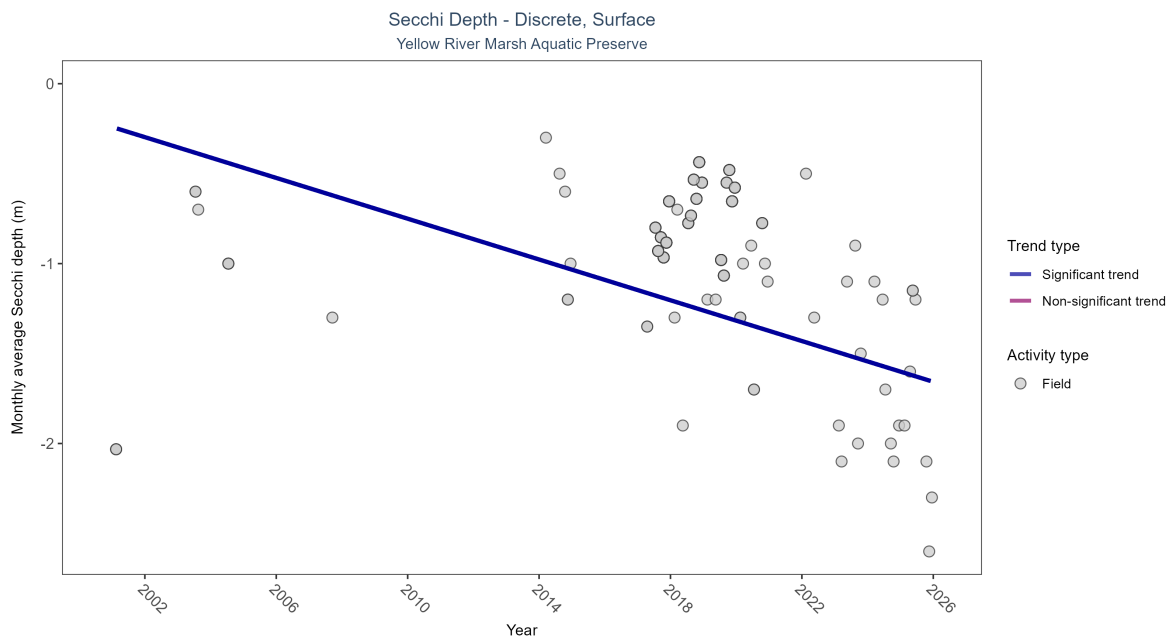


Figure 36: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 19: Seasonal Kendall-Tau Results for - Secchi Depth

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly decreasing trend	246	14	2001 - 2025	-0.8	-0.35722	-0.24043	-0.05667	0.0016

Monthly average Secchi depth became deeper by 0.06 m per year, indicating an increase in water clarity.

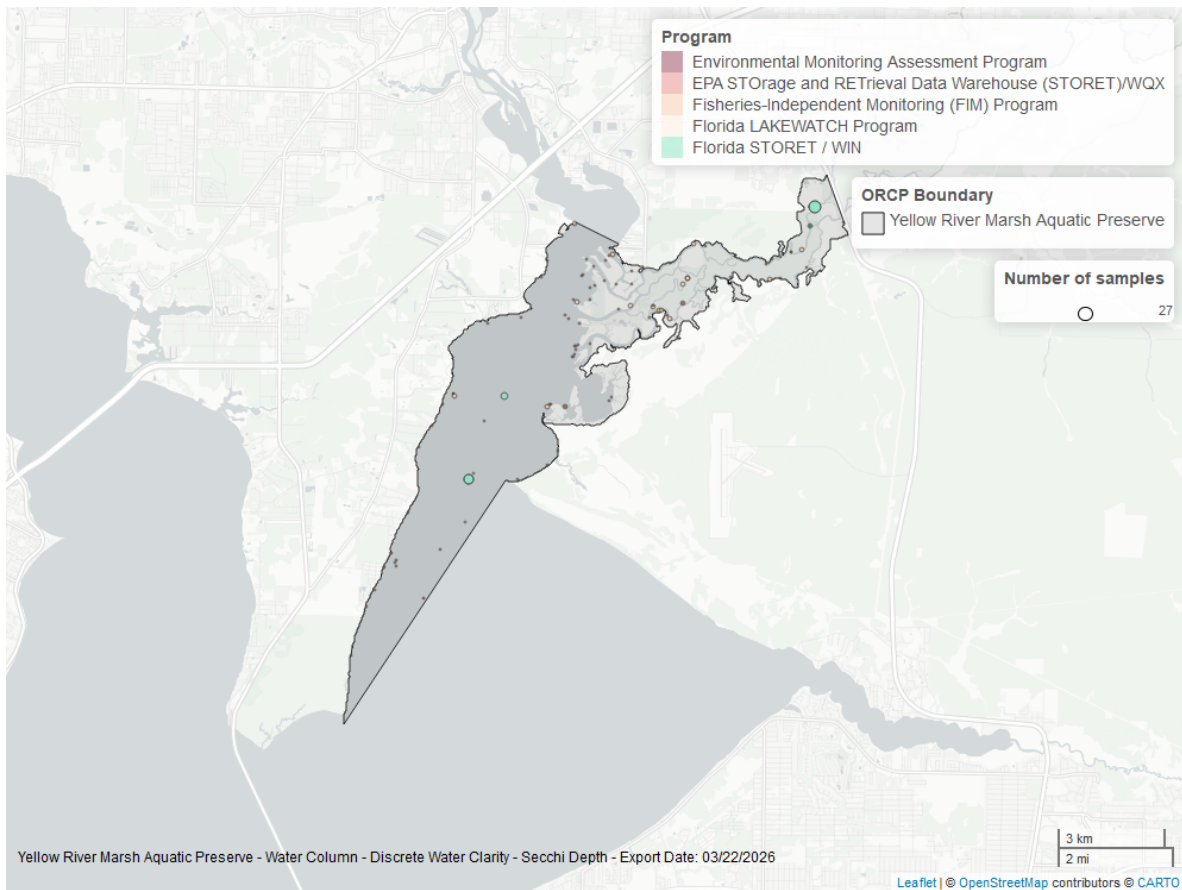


Figure 37: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

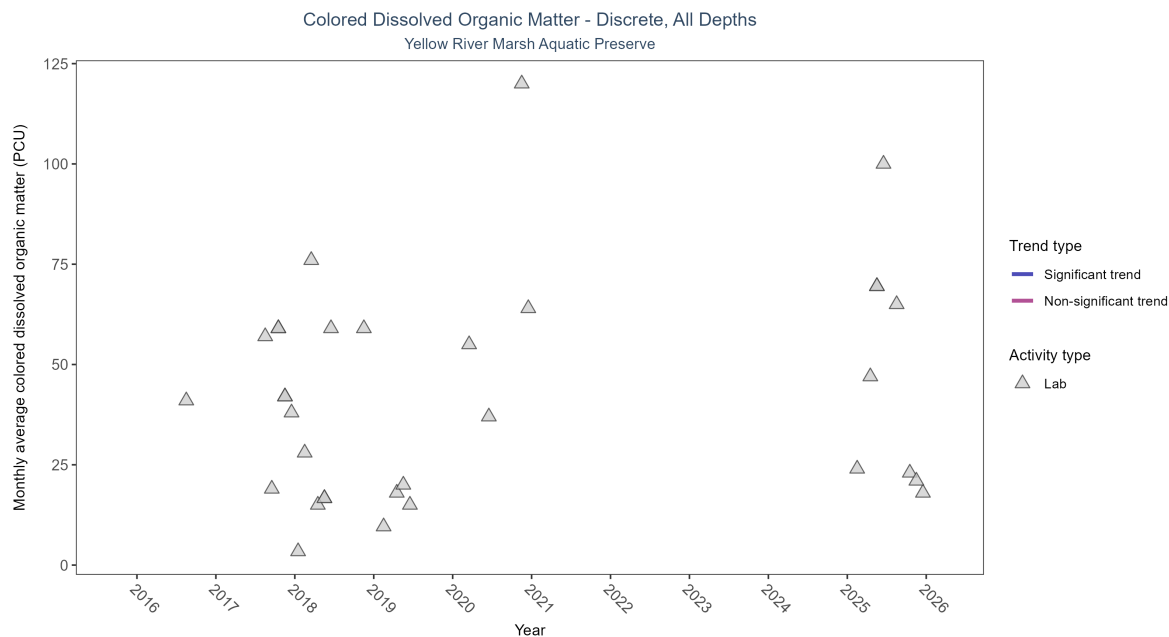


Figure 38: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 20: Seasonal Kendall-Tau Results for - Colored Dissolved Organic Matter

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data to calculate trend	33	6	2016 - 2025	29	-	-	-	-

There was insufficient data to fit a model for colored dissolved organic matter.

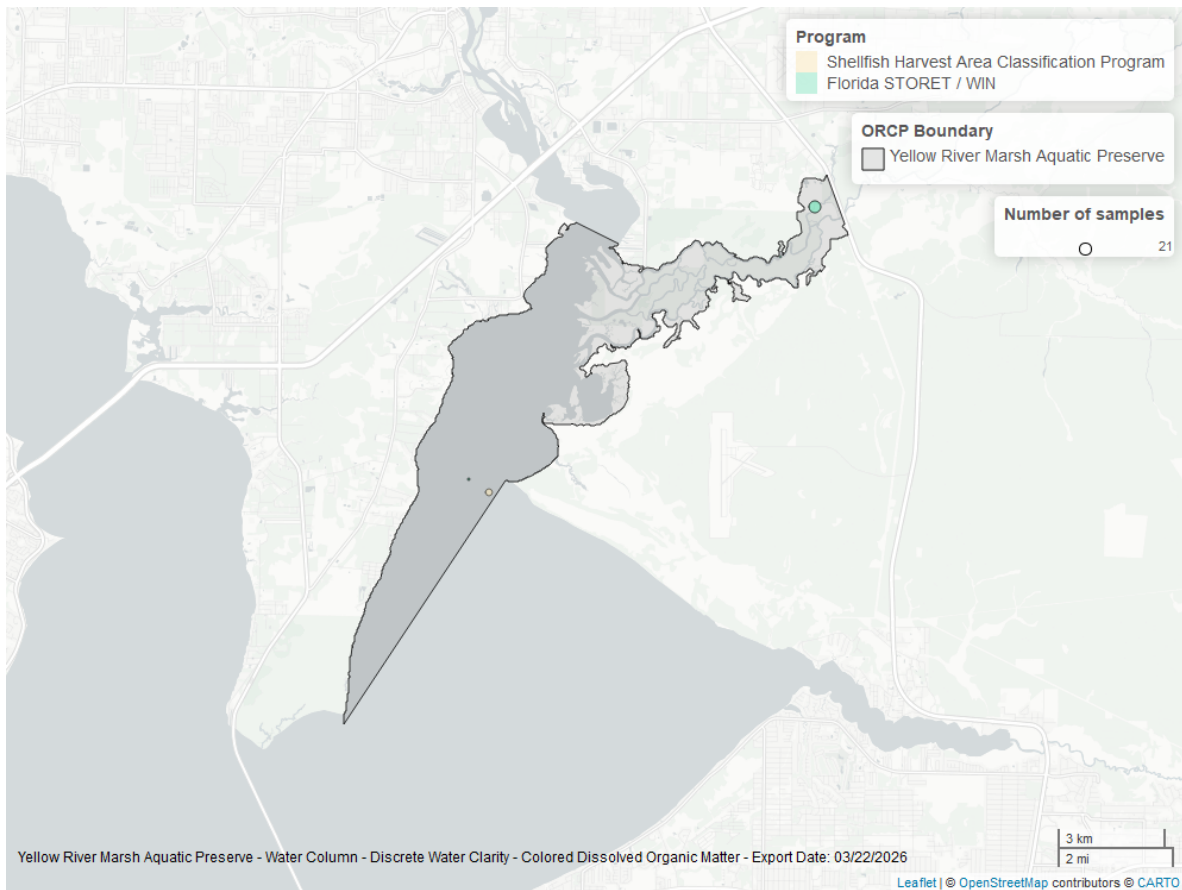


Figure 39: Map showing location of discrete water quality sampling locations within the boundaries of *Yellow River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.